• STRATEGIC INTELLIGENCE BRIEFING / CLASSIFICATION: INTERNAL

# DSCR GODMODE *ULTRAPLAN*

Path to Elite: Mastering the **Six Functions** of a category-defining DSCR lender — borrower filter, lender/investor fit, file cleanliness, and niche clarity. The lender that survives will not be the one with the flashiest ads, but the one with the cleanest operator doctrine.

## THE SIX-FUNCTION DOCTRINE

- 01 Scenario Accuracy — 10-min verdict
- 02 Guideline Intelligence — lender fit
- 03 Borrower Trust — credible quotes
- 04 Capital Partner Trust — clean files
- 05 Distribution — repeatable channels
- 06 Risk Discipline — early decline

| VERSION | DATE | HORIZON | STATUS |
|---|---|---|---|
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CHAPTER 01

# Executive Summary: The Six-Function Doctrine

*The DSCR lender that survives will not be the one with the flashiest ads. It will be the one with the best borrower filter, the best lender/investor relationships, the cleanest files, and the clearest niche.*

This document is the operating doctrine for transforming the DSCR Intelligence Platform from a capable underwriting tool into a category-defining lender intelligence system. The thesis is simple: the residential investment real estate finance market is undergoing a structural reset as of mid-2026, with interest rates holding in the 6.50% to 8.25% band for DSCR product, capital partners tightening guideline matrices, and regulatory scrutiny on prepayment penalties and STR-backed income reaching an all-time high. In this environment, the operators who win are not the ones with the largest marketing budgets; they are the ones who can, within ten minutes of receiving a file, tell the borrower the truth about whether the loan is fundable, name the lender who will fund it, and produce a clean submission that closes on the first pass.

The platform's purpose is to make that ten-minute verdict mechanically reproducible. It does so by separating the lending question from the survival question — Track A asks whether the loan meets a lender or capital partner's qualification matrix; Track B asks whether the investor should do the deal at all once real-world losses, shocks, and liquidity risk are accounted for. The two tracks never mix. They are reported side-by-side so the operator, the borrower, and the capital partner each see the same truth from their own vantage point. This dual-track discipline is what separates a credible DSCR shop from a broker that quotes rate sheets and prays.

Beyond the dual-track core, the platform must master six functions to reach elite operating status. Each function has an elite standard — a measurable performance bar that, when met, places the operator in the top decile of DSCR originators. The Six-Function Doctrine is the organizing principle of this Ultraplan: every upgrade, every code module, every operational change must trace back to one of the six functions, and any capability that does not serve at least one of them is a distraction to be deprioritized.

## The Six-Function Doctrine at a Glance

| # | Function | Elite Standard | Platform Module |
|---|---|---|---|
| 01 | Scenario Accuracy | GO / NO-GO verdict with confidence score in under 10 minutes | engine.ts + preflightGate.ts + rentCompAggregator.ts |
| 02 | Guideline Intelligence | 25+ verified lenders with auto-fit scoring and two-quote rule | lenders.ts + lenderGuidelines.ts + fitScorer.ts |
| 03 | Borrower Trust | Every quote regulator-ready, backed by full constraint disclosure | quoteExplainer.ts + pdfQuotePack.ts |
| 04 | Capital Partner Trust | Zero-defect file standard, first-pass clean rate above 90% | fileCompletenessEngine.ts + defectScorer.ts |
| 05 | Distribution | 60%+ of revenue from repeat referral channels, not random leads | referralPortal.ts + channelAttribution.ts |
| 06 | Risk Discipline | Hard decline gates + adverse-action compliance, false-decline below 5% | declineGate.ts + adverseActionEngine.ts |

## Why Doctrine, Not Features

A common failure mode in DSCR shops is to chase feature breadth — more lenders, more calculators, more dashboards — without ever raising the operator's decision quality. The result is a tool that produces more quotes but not more closings, because the underlying judgment is unchanged. The Six-Function Doctrine inverts this: each function is defined by an external outcome (a clean closing, a declined bad file, a repeat referral), not by an internal feature count. The platform's job is to make the operator's judgment mechanically reproducible across scenarios, lenders, and borrowers, so that the shop's hit rate, turn time, and unit economics improve quarter over quarter regardless of who is on the desk.

This doctrinal framing has three consequences for how the Ultraplan is structured. First, every chapter that follows ties a specific function to a measurable standard, a set of upgrades, a code module inventory, and a KPI; chapters that do not advance at least one function are out of scope. Second, the implementation roadmap in Chapter 10 sequences the work so that foundational correctness (Track A/B engine, payment factor audit, golden test suite) precedes expansion (lender matrix to 25+, distribution portal) — building on a broken core produces more brokenness, not more capability. Third, the success metrics in Chapter 11 are leading indicators (time-to-verdict, first-pass clean rate, false-decline rate) rather than trailing revenue metrics, because leading indicators tell the operator whether the doctrine is being lived before the revenue confirms it.

> **THE IRON RULE OF GODMODE**
> Every feature request, lender addition, UI change, or operational modification must be traceable to exactly one of the Six Functions. If a change serves no function, it is rejected. If a change serves two functions, it is reframed as a cross-cutting capability and tracked in Chapter 9. Doctrine is what prevents the platform from becoming a feature graveyard.

## The Three Audiences of Every Quote

A DSCR quote is read by three audiences with different decision criteria, and the platform must speak legibly to all three simultaneously. The borrower cares whether the deal closes and at what cost of capital; they will judge the quote by its rate, fees, and whether the constraints feel fair and explained. The capital partner — a lender's underwriter, an investor's asset manager, or a credit committee — cares whether the file is clean, complete, and defensible; they will judge the quote by its defect rate, documentation stack, and audit trail. The operator (you) cares whether the ten minutes spent produced a verdict that holds up through closing; you will judge the quote by whether it moved the borrower forward without creating downstream liability. A quote that satisfies only one audience is a failure.

This three-audience model is why Track A and Track B are reported side-by-side and never mixed. Track A speaks to the capital partner's qualification matrix (gross rent divided by PITIA, no vacancy leakage). Track B speaks to the borrower's survival probability (real losses, shocks, liquidity). The operator's role is to translate between the two — to tell the borrower 'yes, this loan qualifies at this lender, but here is what your real cash-on-cash return looks like once vacancy, management, and repair reserves are applied.' That translation is the core value the operator adds; the platform exists to make it mechanically reproducible.

## Reading This Document

Chapter 2 is a candid diagnostic of where the platform stands today against each of the six functions, including the top five capability gaps that block GODMODE status. Chapters 3 through 8 are the function-by-function mastery plans — one chapter per function — each with the same internal structure: elite standard, current gap, required upgrades, new code modules, KPIs, and risks. Chapter 9 covers cross-cutting capabilities (provenance pipeline, regulatory engine, command-center UI, infrastructure) that span multiple functions. Chapter 10 is the 365-day phased roadmap with deliverables, exit criteria, and a risk register. Chapter 11 is the KPI framework. Chapter 12 is the appendix — lender registry, state PPP matrix, data source catalog, and code module inventory.

The document is dense by design. It is intended to be read once cover-to-cover by the operator desk, then used as a reference document during execution. The chapter openers and callout boxes are designed to be scannable in under a minute each, so a reader who only has ten minutes can extract the doctrine, the diagnostic, and the roadmap headline without reading every paragraph.

CHAPTER 02

# Current State Diagnostic

*You cannot fix what you have not measured. This chapter is the honest scorecard of where the platform stands today against each of the Six Functions, including the five most consequential capability gaps.*

The current platform (v7.0 build state) has a defensible core but large capability gaps in the upper-function layers. The Track A / Track B engine, the seven verified lenders, the provenance tag system, and the state PPP scaffolding together constitute a credible underwriting tool — roughly a 2.6 out of 5 average across the six functions. The platform is not yet a category-defining intelligence system; it is a competent calculator with the right architectural bones. The purpose of this diagnostic is to name, without euphemism, what works, what is broken, and what is missing entirely, so that the upgrade roadmap can target the highest-leverage gaps first.

The diagnostic was conducted by scoring each function on a five-point maturity scale (1 = absent, 2 = scaffolded, 3 = functional, 4 = hardened, 5 = elite), then identifying the specific capabilities that would have to be added or fixed to move each function up by one full maturity point. The scores below are calibrated against external benchmarks — what a top-decile DSCR originator can actually do today, not what a vendor marketing page claims — and they reflect the platform's state as of the date on the cover.

## Maturity Matrix — Current vs Target State

| Function | Current Score | Target (12mo) | Critical Gap |
|---|---|---|---|
| 01 Scenario Accuracy | 2.5 / 5 | 4.5 / 5 | No pre-flight gate; rent comps manual; no Monte Carlo overlay |
| 02 Guideline Intelligence | 2.0 / 5 | 4.5 / 5 | Only 7 verified lenders; no guideline diff engine; no fit scoring |
| 03 Borrower Trust | 2.5 / 5 | 4.5 / 5 | No 'why this rate' explainer; no PDF quote packet with provenance |
| 04 Capital Partner Trust | 1.5 / 5 | 4.5 / 5 | No file completeness engine; no defect scoring; no reconciliation |
| 05 Distribution | 1.0 / 5 | 4.0 / 5 | No referral portal; no channel attribution; no investor CRM |
| 06 Risk Discipline | 2.0 / 5 | 4.5 / 5 | No auto-decline gates; no adverse-action engine; no pattern learning |
| AVERAGE | 1.92 / 5 | 4.42 / 5 | — |

## What Works Today (Defend This)

The Track A / Track B engine is the platform's strongest asset. Track A correctly applies the lender-qualification formula — gross rent divided by PITIA, with no vacancy leakage — and correctly solves for the deal-break rate (the interest rate at which Track A DSCR equals the lender's minimum). Track B correctly applies real-world losses, shocks, and liquidity risk to produce a survival verdict. The two tracks are reported separately, which is the discipline most DSCR shops fail to maintain. The payment factor audit (8.25% maps to 0.0075127, not 0.007568; 7.00% maps to 0.006653) is correct, and the golden test suite locks these values in place so future refactors cannot silently break them.

The seven verified lenders — Griffin at 75-85% LTV, Defy at 80%, Easy Street at 82%, Lima One at 76%, New Silver at 72%, Kiavi at 70%, and Deephaven at 65% — represent a credible starting matrix that covers the LTV spectrum from 65% to 85%. Each lender entry carries provenance tags indicating whether the guideline data is primary-sourced (from the lender's own published matrix), secondary-sourced (from a correspondent or trade publication), or unverified. The state PPP scaffolding correctly identifies the high-stakes jurisdictions (Minnesota prohibited, New Jersey individual prohibited, Illinois restricted, Ohio and Pennsylvania with indexed thresholds, Mississippi with the 5-4-3-2-1 step-down, Washington with an ARM ban that requires secondary verification). These are the right bones; what is missing is the muscle and nervous system layered on top of them.

## What Is Broken (Fix This First)

Three things are broken and must be fixed before any new capability is added. First, the state PPP engine is scaffolded but not operational — the Washington ARM ban is tagged UNVERIFIED, and the Ohio and Pennsylvania indexed thresholds need annual recalibration logic that does not exist yet. Second, the lender matrix lacks any diff engine, meaning that when a lender updates a guideline (raises their DSCR minimum from 1.00 to 1.10, tightens their FICO floor from 660 to 680, or exits a state), the platform has no way to detect the change and flag affected pipeline files. Third, the reserve engine produces a single point estimate rather than the three-scenario output (base / stress / severe) that capital partners actually want to see, and it lacks the geographic coverage overlay that would let a multi-property portfolio reserve be modeled correctly.

> **THE FIVE MOST CONSEQUENTIAL CAPABILITY GAPS**
>
> **Gap 1 — Pre-flight gate.** The platform cannot, in under two minutes, tell the operator whether a file is even worth underwriting (STR legality, zoning, HOA, flood, insurability, title red flags). **Gap 2 — Lender matrix depth.** Seven lenders is half the minimum needed to enforce the two-quote rule across the full file-shape spectrum. **Gap 3 — File completeness engine.** No automated check that the document stack matches the target lender's exact requirements, so first-pass clean rate is unmeasured and likely below 50%. **Gap 4 — Distribution layer.** The platform has no intake portal for repeat investors, agents, or wholesalers, meaning every lead is a random lead. **Gap 5 — Decline pattern learning.** No system captures why files are declined and feeds that back into lender fit scoring, so the same bad file shapes recur.

## What Is Missing Entirely (Build This)

Beyond what is broken, five capabilities are absent entirely and must be built from scratch. The first is the rent comp aggregator — there is no automated pull from RentCast, Rentometer, AirDNA, or MLS 1007, so rent figures are operator-entered and unverified. The second is the document OCR and reconciliation engine — there is no system that reads a rent roll, a lease, and a 1007 market rent opinion and flags inconsistencies before submission. The third is the adverse action letter generator — without it, the platform cannot produce ECOA / Reg B compliant decline notices, which is a compliance exposure. The fourth is the channel attribution system — without it, the shop cannot measure which referral channels produce closed loans versus wasted underwriting hours. The fifth is the Monte Carlo survivorship overlay — without it, Track B produces a single point estimate of investor survival probability, which understates tail risk.

These five missing capabilities, together with the three broken capabilities and the five consequential gaps named in the callout above, define the work. The 365-day roadmap in Chapter 10 sequences this work into four phases: Phase 1 fixes the broken core and adds the pre-flight gate; Phase 2 expands the lender matrix and builds the file completeness engine; Phase 3 stands up the distribution layer and the risk discipline automation; Phase 4 completes the regulatory engine, adds the Monte Carlo overlay, and reaches full GODMODE operating status. The roadmap is intentionally aggressive — twelve months is the window before the market resets again, and the platform must be elite before that reset, not after.

CHAPTER 03

# Function 1 — Scenario Accuracy: The 10-Minute Verdict

*Elite standard: from a property address, you can produce a GO / NO-GO verdict with a confidence score in under ten minutes. Anything slower is a loss of optionality; anything less confident is a guess.*

## Elite Standard

The elite operator can, in under ten minutes from receiving a property address and a basic borrower brief, produce a verdict with three components: a Track A DSCR range with lender-attribution (which two lenders will quote, at what rate band, with what fit tier); a Track B survival verdict (does the deal survive real-world losses, a one-shock event, and a liquidity squeeze); and a confidence score that calibrates how much weight to put on the verdict. The verdict must be wrong less than 5% of the time at the GO/NO-GO threshold, and the confidence score must be calibrated — meaning a 90% confidence verdict is actually correct 90% of the time, not 75% of the time. Calibration is harder than accuracy; an uncalibrated confident verdict is more dangerous than a calibrated uncertain one.

Ten minutes is not an arbitrary target. It is the time within which a borrower on the phone will stay engaged, the time within which an agent referral will not lose patience, and the time within which a wholesaler's file can be triaged before it competes for the operator's attention with five other files. Beyond ten minutes, the operator has lost the moment of peak borrower trust; beyond twenty minutes, the file has typically been shopped to a competitor. The platform's job is to compress the ten-minute verdict into a reproducible workflow that any trained operator can execute, not just the founder.

| 10 min | 95% | 90% | 5% |
|---|---|---|---|
| TIME-TO-VERDICT SLA | VERDICTS UNDER 10 MIN | CONFIDENCE CALIBRATION | FALSE GO RATE CEILING |

## Current Gap

Today the platform produces a verdict in roughly 25 to 45 minutes for a clean file, and significantly longer for a file with STR income, mixed-use zoning, or a non-standard entity structure. The bottlenecks are: (a) rent figures are operator-entered after manual lookup on RentCast, Rentometer, and AirDNA, with no automated aggregation or confidence weighting; (b) insurance and tax estimates require separate manual lookups that often consume five to ten minutes per file; (c) the pre-flight gate does not exist, so files enter the underwriting queue that should have been declined at intake (HOA STR ban, flood zone AE without HOI, prohibited state); (d) there is no Monte Carlo overlay, so Track B produces a single-point survival verdict that understates tail risk and forces the operator to manually stress-test.

## Required Upgrades

### 1. Instant Pre-flight Gate

The pre-flight gate is the single highest-leverage upgrade in the entire Ultraplan. It runs in under 90 seconds from a property address and produces a red / amber / green verdict on seven dimensions: STR legality (CLEAR / RESTRICTED / UNCERTAIN / PROHIBITED), zoning classification, HOA STR restrictions, flood zone (with HOI implication), insurability (Carrier of last resort flagged), title red flags (recent transfer, liens, judgment), and property type eligibility (SFR / 2-4 / condo / mixed / commercial). Red on any dimension auto-declines the file before underwriting time is spent; amber routes to manual review with a specific reason code; green proceeds to full underwriting. The gate eliminates 20-30% of bad files at intake, which is the largest single productivity gain available.

### 2. Automated Rent Comp Aggregator

The rent comp aggregator pulls from four sources in parallel — RentCast (long-term market rent), Rentometer (long-term percentile range), AirDNA (STR projected revenue with 20% Track-A discount applied), and MLS 1007 (when an appraisal-style market rent opinion is available) — and produces a triangulated rent figure with a confidence weight. The lower-of rule (qualifying rent = minimum of lease rent and 1007 market rent) is enforced automatically. The three STR income worlds (long-term market rent, forecast STR, historical STR) are kept strictly separated and never mixed in the same calculation. This eliminates the single largest source of operator-entered error and saves 8-12 minutes per file.

### 3. Lender Matrix Auto-Router

Given the file shape (LTV requested, FICO, property type, state, loan amount, DSCR signal from the rent comp), the auto-router returns the two lenders most likely to quote (one flex lender, one rate-competitive lender, per the two-quote rule) and the rate band each will quote at. This is the operational manifestation of Function 2 (Guideline Intelligence) — the router depends on the lender matrix being complete and current. The router also surfaces the fit tier for each lender (Strong / Standard / Conditional / Unlikely / Does Not Meet) so the operator can immediately see whether to push for a quote or to triage.

### 4. Confidence-Weighted DSCR Range Output

Instead of producing a single point estimate of Track A DSCR, the platform produces a range with a confidence interval: e.g., 'Track A DSCR 1.05 to 1.12, 90% confidence, driven by rent comp uncertainty of ±$75/month.' The confidence score combines four components: source timeliness (40% weight), source quantity (25%), source quality (25%), and cross-source consistency (10%). A verdict with sub-70% confidence is flagged for manual verification before it is shared with the borrower. This is the calibration discipline that separates elite operators from confident-but-wrong ones.

### 5. Enhanced Deal-Break Rate Solver

The deal-break rate solver already exists and correctly identifies the interest rate at which Track A DSCR equals the lender's minimum (the flagship transaction solves to 7.67%). The enhancement is to produce a deal-break curve, not a single point — showing how the deal-break rate shifts as LTV moves from 70% to 80%, as the rent comp moves from -5% to +5%, and as the DSCR minimum moves from 1.00 to 1.20. The curve lets the operator see whether the deal is fragile (deal-break rate within 50bps of market) or robust (deal-break rate 200bps above market) before recommending a lender.

### 6. Monte Carlo Survivorship Overlay

Track B currently produces a single-point survival verdict. The Monte Carlo overlay runs 10,000 trials varying vacancy (5-15%), repair shocks (probability and severity distributions by property age and type), management fee inflation, interest rate reset risk (for ARM and IO loans), and exit liquidity (refinance probability as a function of equity buildup). The output is a survivorship probability at 12, 24, and 36 months — the probability that the investor can carry the property without injecting additional capital. A deal with 95% 12-month survival but 55% 36-month survival is fundamentally different from one with 90% / 88% / 85%, even though the 12-month numbers look similar; the overlay surfaces this distinction.

### 7. Tax and Insurance Estimator Integrations

Property tax estimates are pulled from county assessor APIs (where available) or modeled from assessed value plus millage rate; insurance estimates are pulled from a HOI estimation service (Hippo, The Zebra, or similar) with a carrier-of-last-resort flag for properties in windpool or wildfire zones. These integrations eliminate 5-10 minutes of manual lookup per file and dramatically reduce the variance in PITIA estimation that currently makes Track A DSCR noisy.

## New Code Modules

| Module | Purpose | Dependencies | Est. LoC |
|---|---|---|---|
| preflightGate.ts | 7-dimension pre-flight gate, red/amber/green output | zoningApi, floodApi, strLegalityDb, hoaDb | ~650 |
| rentCompAggregator.ts | 4-source rent comp pull, triangulation, confidence | RentCast, Rentometer, AirDNA, MLS 1007 | ~820 |
| insuranceEstimator.ts | HOI estimate + carrier-of-last-resort flag | HOI estimation API, windpool/wildfire zones | ~430 |
| taxEstimator.ts | County assessor pull or millage model | County assessor APIs, millage db | ~380 |
| confidenceScorer.ts | 4-component confidence score (time/qty/qual/consis) | Provenance tags on all inputs | ~290 |
| dealBreakCurve.ts | Deal-break rate curve across LTV/rent/DSCR sweep | engine.ts (existing) | ~340 |
| monteCarloSurvivorship.ts | 10,000-trial Track B survival overlay | engine.ts, reserveEngine.ts | ~720 |

## Workflow Diagram — The 10-Minute Verdict

The compressed workflow is: **T+0:00** Operator enters address + borrower brief. **T+0:30** Pre-flight gate returns red/amber/green on 7 dimensions. **T+1:30** If green: rent comp aggregator returns triangulated rent with confidence score. **T+3:00** Tax and insurance estimators return PITIA components. **T+4:00** Track A DSCR range computed; lender auto-router returns two fit-tier lenders with rate bands. **T+5:30** Track B survival verdict with Monte Carlo overlay returns 12/24/36-month survivorship. **T+7:00** Deal-break curve computed; verdict and quote packet drafted. **T+9:30** Confidence score finalized; if ≥85% confidence, verdict delivered to borrower with PDF quote packet; if <85%, manual review flag and re-pull of weakest input. The workflow is designed to leave 30 seconds of operator judgment time at the end, which is the irreducible human component.

## Key Performance Indicators

| KPI | Baseline (today) | 90-day target | 365-day target |
|---|---|---|---|
| Median time-to-verdict | 25-45 min | 12 min | <10 min |
| % verdicts under 10 min | ~20% | 60% | 95% |
| Confidence calibration (Brier) | n/a | 0.18 | <0.10 |
| False GO rate (deals that fail) | unmeasured | 12% | <5% |
| Pre-flight gate auto-decline rate | 0% | 20% | 28% |
| Rent comp manual override rate | 100% | 40% | <15% |

## Risks and Mitigations

The principal risk is data-source reliability. RentCast and Rentometer have known accuracy gaps in secondary and tertiary markets, and AirDNA STR projections degrade quickly in markets with thin short-term rental inventory. Mitigation: the confidence score must surface low-confidence verdicts for manual verification, and the rent comp aggregator must always display the spread between sources so the operator can spot disagreement. A secondary risk is API rate limits and cost — RentCast and AirDNA both charge per-pull, and an uncontrolled aggregator could run up significant monthly cost. Mitigation: aggressive caching (rent comps for a given address are valid for 30 days), bulk-pull batching for portfolio scenarios, and a monthly cost ceiling that throttles pulls when hit. A tertiary risk is false-decline from the pre-flight gate — a property flagged PROHIBITED for STR when it is in fact legal under a recent ordinance change. Mitigation: the STR legality database must have a weekly refresh cadence and an operator-override path with audit logging.

CHAPTER 04

# Function 2 — Guideline Intelligence: Lender Fit

*Elite standard: a real-time guideline database covering 25+ verified lenders, with fit scoring that auto-maps each file to the right two lenders and a diff engine that surfaces every guideline change before it kills a pipeline file.*

## Elite Standard

The elite operator knows which lender fits which file before they pick up the phone. The platform's job is to make this knowledge mechanical: given a file shape, return the two lenders most likely to quote (one flex lender for borderline files, one rate-competitive lender for clean files), with a Fit Tier (Strong / Standard / Conditional / Unlikely / Does Not Meet) for each, a rate band that the lender is currently quoting for that file shape, and a list of any guideline changes in the last 30 days that affect the file. The two-quote rule is non-negotiable: every quoted file goes out with two lender options so the borrower sees real competition, not a single take-it-or-leave-it offer.

Beyond the per-file fit, the elite operator maintains a lender relationship tier system that tracks which lenders actually close the files sent to them (A-tier relationships close >70% of submitted files; B-tier 40-70%; C-tier <40%). This relationship data feeds back into the fit scoring — a lender may have guidelines that fit a file perfectly but a relationship tier that says they will not close it, in which case the platform recommends the next-best lender with a stronger relationship. This is the difference between a fit score that says 'this loan meets the guideline' and one that says 'this loan will close.'

| 25+ | 100% | >85% | <7d |
|---|---|---|---|
| VERIFIED LENDERS | TWO-QUOTE COMPLIANCE | FIT ACCURACY (WILL-CLOSE) | GUIDELINE DIFF LATENCY |

## Current Gap

The current lender matrix has seven verified lenders — Griffin, Defy, Easy Street, Lima One, New Silver, Kiavi, and Deephaven. This is roughly half the minimum needed to enforce the two-quote rule across the full file-shape spectrum. Seven lenders can cover either the LTV spectrum (65-85%) or the FICO spectrum (660-760) adequately, but not both, and there is no coverage for niche file shapes (condotels, non-warrantable condos, mixed-use, log homes, foreign national borrowers). Beyond depth, the matrix has three structural deficits: there is no guideline diff engine (when a lender updates a guideline, no one knows until a file gets declined); there is no fit scoring (the operator eyeballs which lender to send to); and there is no relationship tier tracking (the platform does not know which lenders actually close the files sent to them).

## Required Upgrades

### 1. Expand Lender Matrix to 25+ Verified Lenders

The target lender matrix covers the full file-shape spectrum with at least three lenders in each LTV band (65-70%, 70-75%, 75-80%, 80-85%) and at least two in each property type category (SFR, 2-4 unit, condo, condotel, mixed-use, manufactured). The expansion prioritizes lenders that fill current matrix gaps — Visio Lending (75-80%

LTV, strong on 2-4 unit), LendingOne (70-80% LTV, strong on condotels), RCN Capital (65-75% LTV, strong on mixed-use), Anchor Loans (70-80% LTV, strong on fix-and-flip bridge), Lima One expanded programs (75% LTV on STR), Velocity Mortgage (65-80% LTV, strong on manufactured), and others. Every lender entry carries provenance tags and a verification date; entries older than 90 days are flagged for re-verification.

### 2. Guideline Diff Engine

The guideline diff engine monitors each lender's published guideline matrix on a weekly cadence and surfaces any change within 7 days of publication. A change is classified as material (DSCR minimum, LTV cap, FICO floor, state exclusion added, program exit) or cosmetic (formatting, contact info, marketing copy). Material changes trigger a pipeline impact analysis: every file in the pipeline that was routed to that lender is flagged, the operator is notified, and an alternative lender is recommended if the file no longer fits. This eliminates the silent killer of DSCR pipelines — the lender that tightens a guideline mid-pipeline and declines a file that would have closed three weeks earlier.

### 3. Fit Tier Auto-Mapping

Given a file shape, the fit scorer returns a Fit Tier for each lender in the matrix: **Strong** (file meets every guideline with margin), **Standard** (file meets every guideline), **Conditional** (file meets guidelines but with a borderline dimension — DSCR within 0.05, FICO within 20 points, LTV within 2%), **Unlikely** (file misses one guideline), **Does Not Meet** (file misses multiple guidelines). The auto-router then enforces the two-quote rule: pick one Strong or Standard lender (rate-competitive) and one Conditional or Unlikely lender (flex), and present both to the borrower with the fit tier visible so the borrower understands the relative confidence of each quote.

### 4. Two-Quote Rule Enforcement

The two-quote rule is a hard platform constraint: no quote leaves the system with only one lender attached. The rule exists because a single-lender quote is not a quote, it is a take-it-or-leave-it offer, and borrowers correctly distrust such offers. The enforcement is simple: the quote packet generator refuses to render a PDF with fewer than two lender sections. The operator can override the rule (for example, when only one lender in the matrix fits a niche file shape) but the override is logged and reviewed monthly. Repeated overrides trigger a matrix gap analysis — if the same niche keeps producing single-lender quotes, a lender must be added to the matrix to fill the gap.

### 5. Lender Relationship Tier Tracker

Every submitted file carries an outcome tag (closed / declined / withdrawn / approved-but-not-funded) and the outcome is recorded against the submitting lender. The relationship tier is computed quarterly: A-tier (closes >70% of submitted files), B-tier (40-70%), C-tier (<40%). The fit scorer weights Fit Tier by Relationship Tier — a lender with a Strong Fit Tier but a C-tier relationship is demoted in the auto-router recommendation, because a perfect guideline match with a lender that will not close is worth less than a Standard match with an A-tier relationship. This is the platform's feedback loop for reality.

### 6. Live Pricing Grid Integration

Where lenders expose a live pricing grid (Optimal Blue, PPE, or lender-direct APIs), the platform pulls the current rate sheet for the file shape and returns the actual quote — not a modeled rate band. Live pricing is preferred over modeled pricing whenever available, and the quote packet notes whether the rate is live or modeled. For lenders without live pricing, the modeled rate band uses the lender's last-known grid plus a market adjustment factor (a function of the 10-year Treasury and MBS spread) to estimate current pricing. Modeled quotes carry a disclaimer and a refresh cadence (24 hours).

### 7. Decline-Pattern Learning

Every lender decline is tagged with the lender's stated reason code (DSCR, FICO, LTV, property type, state, entity, reserves, etc.) and recorded in a decline-pattern database. The database is mined quarterly for patterns: 'Lender X declines 60% of files with DSCR 1.00-1.05 even though their published minimum is 1.00' or 'Lender Y declines 80% of condotels even though they are not on the exclusion list.' These patterns feed back into the fit scorer as adjustments — a lender's effective DSCR minimum may be 1.10 even though their published minimum is 1.00, and the fit scorer uses the effective number. This is how the platform learns what lenders actually do, not what they say they do.

## Target Lender Expansion List (Year 1)

| Lender | LTV Band | Niche Strength | Verification Status |
|---|---|---|---|
| Visio Lending | 75-80% | 2-4 unit, STR | Target — Q2 |
| LendingOne | 70-80% | Condotel, foreign national | Target — Q2 |
| RCN Capital | 65-75% | Mixed-use, bridge | Target — Q2 |
| Anchor Loans | 70-80% | Fix-and-flip, bridge | Target — Q3 |
| Velocity Mortgage | 65-80% | Manufactured, rural | Target — Q3 |
| Visions FCU (correspondent) | 75-80% | Rate-competitive owner-occ adjacent | Target — Q3 |
| Park Place Finance | 70-80% | Cash-out, no-seasoning | Target — Q3 |
| Anchor Six Lending | 70-80% | STR, condotels | Target — Q4 |
| Broadmark Realty | 65-75% | Bridge, fix-flip | Target — Q4 |
| CoreVest Finance | 70-80% | Portfolio, single-asset | Target — Q4 |
| Lima One STR program | 75% | STR expanded | Verify — Q1 |
| Griffin expanded STR | 75-80% | STR with historical income | Verify — Q1 |
| Marquee Funding | 65-75% | CA specialty, jumbo | Target — Q4 |
| Pacwest Capital | 70-80% | West Coast, jumbo DSCR | Target — Q4 |
| Grand Coast Financial | 65-75% | Bridge, multifamily | Target — Q4 |
| Pinnacle DSCR | 70-80% | Rate-competitive, SFR | Target — Q4 |
| Stratton Equities | 70-80% | No-income, alternative | Target — Q4 |
| Centerpoint Lending | 70-80% | SFR, 2-4 unit | Target — Q4 |

## New Code Modules

| Module | Purpose | Est. LoC |
|---|---|---|
| lenderGuidelines.ts | Structured guideline JSON per lender (25+ entries) | ~2,400 |
| guidelineDiff.ts | Weekly diff + material change classifier + pipeline impact | ~640 |

| Module | Purpose | Est. LoC |
|---|---|---|
| fitScorer.ts | File shape -> Fit Tier per lender, with relationship tier adjustment | ~520 |
| pricingGridFeed.ts | Optimal Blue / PPE / lender-direct live pricing integration | ~880 |
| relationshipTracker.ts | Outcome tagging, quarterly tier computation, history | ~430 |
| declinePatternLearner.ts | Reason-code mining, effective-minimum inference | ~570 |
| twoQuoteEnforcer.ts | Quote packet refuses to render with <2 lenders; override log | ~210 |

## Key Performance Indicators

| KPI | Baseline | 90-day | 365-day |
|---|---|---|---|
| Verified lenders in matrix | 7 | 14 | 25+ |
| Two-quote rule compliance | ~60% | 95% | 100% |
| Fit accuracy (will-close prediction) | unmeasured | 70% | >85% |
| Guideline diff detection latency | manual | <14 days | <7 days |
| A-tier relationships | 0 tracked | 5 | 12+ |
| Pipeline files lost to guideline change | unmeasured | <8% | <3% |

## Risks and Mitigations

The principal risk is guideline data accuracy. Lender matrices change frequently (some lenders update monthly, some quarterly, some ad hoc), and stale data is worse than no data because it produces confident wrong answers. Mitigation: every guideline entry carries a verification date and a re-verification cadence (90 days for stable lenders, 30 days for volatile ones), and entries past their re-verification date are flagged in the fit scorer with a confidence penalty. A secondary risk is relationship data integrity — the relationship tier is only as good as the outcome tagging, and operators may forget to tag outcomes. Mitigation: outcome tagging is a required field at file close-out, enforced by the workflow system; files without outcome tags appear on a daily exception report. A tertiary risk is overfitting — the decline-pattern learner may infer an effective minimum from a small sample (e.g., 3 declines out of 4 submissions) and over-penalize a lender. Mitigation: effective minimums require a minimum sample size (10 submissions) before they override the published minimum, and the inference is logged for operator review.

CHAPTER 05

# Function 3 — Borrower Trust: Quote Credibility

*Elite standard: investors believe your quote because you explain the constraints. Every quote is regulator-ready, backed by full disclosure, and the borrower can see exactly why this rate, this LTV, this term.*

## Elite Standard

Borrower trust is not a function of having the lowest rate; it is a function of being the most credible. The elite operator produces a quote that the borrower believes because the quote explains itself — the rate is decomposed into base rate, adjustment matrix, and DSCR bridge; the constraints are visualized so the borrower can see what limits the LTV, DSCR, and term; and the borrower can toggle scenarios ('what if rate drops 50bps', 'what if I put down 5% more') and see the answer update in real time. The quote packet is a PDF with provenance tags on every number, so the borrower — and their CPA, attorney, or partner — can audit the assumptions. When the borrower asks 'why is this rate 8.25% and not 7.5%?', the operator has a one-click answer.

Trust is also a function of honesty about the constraints. The elite quote says 'this loan qualifies at 75% LTV, not 80%, because the DSCR at 80% is 0.96 and the lender minimum is 1.00. If you can document an additional $200/month in rent, the DSCR at 80% becomes 1.02 and we can re-quote at the higher LTV.' This honesty converts the quote from a take-it-or-leave-it offer into a collaborative problem-solving conversation, which is the foundation of repeat business. Borrowers do not return to operators who quote low and decline late; they return to operators who quote honest and close on time.

| >70% | >50 | <2d | 100% |
|---|---|---|---|
| QUOTE ACCEPTANCE RATE | BORROWER NPS | TIME-TO-TRUST | QUOTES WITH PROVENANCE |

## Current Gap

Today the platform produces a rate, a payment, and a Track A / Track B verdict, but it does not explain itself. The borrower sees a number; they do not see why the number is what it is, what would change it, or what constraints limit it. The quote is delivered verbally or in a free-form email, with no provenance, no scenario toggles, and no audit trail. The result is that borrowers shop the quote — they take the number to a competitor and ask 'can you beat this?', because they have no reason to believe the number is the right number. This is the principal reason operators lose deals they should close: not because the quote is wrong, but because the quote is not credible.

## Required Upgrades

### 1. 'Why This Rate' Explainer

The explainer decomposes the quoted rate into three components: (a) the lender's base rate for the file shape (the rate before adjustments, looked up from the live pricing grid or modeled from the last-known grid); (b) the adjustment matrix (each adjustment that applies — LTV adjustment, FICO adjustment, DSCR adjustment, property type adjustment, occupancy adjustment, reserve adjustment — with the basis-point impact of each); (c) the DSCR bridge (gross rent to PITIA build-up, with each component of PITIA broken out). The output is a visual waterfall that starts at the base rate and ends at the quoted rate, with each adjustment as a labeled step. The borrower can see, in 30 seconds, exactly why 8.25% and not 7.5%.

### 2. Constraint Visualization

The constraint visualization shows what is binding the LTV, the DSCR, and the term. For LTV: 'Lender caps LTV at 80% for this property type; you requested 80%, so LTV is at the cap.' For DSCR: 'At 80% LTV, PITIA is $2,270/month; qualifying rent at $2,300 gives DSCR 1.01; lender minimum is 1.00, so DSCR is the binding constraint.' For term: 'Lender offers 30/25/20/15 year terms; 30-year produces the lowest PITIA and the highest DSCR, so 30-year is recommended.' The visualization also shows what would unbind each constraint — 'If you put down 5% more (75% LTV), PITIA drops to $2,130 and DSCR rises to 1.08, which moves the file from Conditional to Strong Fit Tier.'

### 3. Side-by-Side Lender Comparison with Reason Codes

The two-quote rule produces two lender quotes; the side-by-side comparison shows them next to each other with reason codes for the differences. Reason codes are short, specific, and operator-meaningful: 'RATE-001: Lender A has a 25bps pricing adjustment for 75% LTV that Lender B does not'; 'FICO-003: Lender A requires 680 FICO minimum, Lender B requires 700, your 705 FICO qualifies for both but is closer to Lender A's floor'; 'PPP-002: Lender A has a 5-year soft prepay, Lender B has a 3-year yield-maintenance, both are permissible in this state but Lender B's penalty is heavier if you sell in year 2.' The reason codes make the comparison legible — the borrower can see why the two quotes are different and choose on the merits.

### 4. 'What Would Change This' Scenario Toggles

The quote packet includes five interactive scenario toggles that recompute the quote in real time: (a) rate +/- 50bps; (b) DSCR +/- 0.10; (c) LTV +/- 5%; (d) term 30/25/20/15-year; (e) rent comp +/- $100/month. Each toggle shows the new PITIA, the new DSCR, the new fit tier, and whether the lender would still quote at the new file shape. The toggles let the borrower explore their own what-ifs without requiring operator time — they can see for themselves that 'yes, putting down 5% more gets us to 80% LTV at 7.75% instead of 75% LTV at 8.00%' without picking up the phone. This is the principal trust-builder: the platform lets the borrower audit the operator's reasoning.

### 5. PDF Quote Packet with Provenance Tags

The quote packet is a 4-6 page PDF that includes: (a) cover page with file summary, lender A vs lender B comparison, and quote expiration; (b) 'Why this rate' waterfall for each lender; (c) constraint visualization; (d) Track A and Track B verdicts with confidence scores; (e) scenario toggle results table; (f) assumptions appendix with provenance tags on every input ([VERIFIED-Primary], [VERIFIED-Secondary], [UNVERIFIED]) and a refresh date for each. The PDF is delivered to the borrower within 30 minutes of the verdict; it is the artifact that the borrower will show to their CPA, attorney, and partners, so it must be defensible. The provenance tags are non-negotiable: a quote without provenance is an opinion; a quote with provenance is an audit.

### 6. Audit Log of Every Assumption and Override

Every assumption (rent comp figure, insurance estimate, tax estimate, DSCR minimum, rate grid version) is logged with the source, the timestamp, and the operator who entered or approved it. Every override (operator manually adjusting the rent comp, operator overriding the auto-router's lender recommendation, operator overriding the pre-flight gate) is logged with the operator, the timestamp, and a free-text reason. The audit log serves three purposes: it makes the operator's reasoning reviewable, it makes the platform's recommendations improvable (patterns of override reveal where the platform is wrong), and it makes the file defensible in the event of a borrower complaint or a regulatory inquiry.

## New Code Modules

| Module | Purpose | Est. LoC |
|---|---|---|
| quoteExplainer.ts | 'Why this rate' waterfall + adjustment matrix | ~680 |
| rateAdjustmentEngine.ts | Base rate lookup + adjustment application | ~540 |
| constraintVisualizer.ts | LTV/DSCR/term binding constraint surfacing | ~470 |
| scenarioToggles.ts | 5-toggle interactive what-if engine | ~390 |
| pdfQuotePack.ts | 6-page PDF generator with provenance tags | ~1,100 |
| reasonCodes.ts | Standardized reason code library (~50 codes) | ~320 |
| auditLog.ts | Immutable log of every assumption and override | ~430 |
| borrowerPortal.ts | Web UI for borrowers to view and toggle quotes | ~1,800 |

## Quote Packet Layout (6 pages)

| Page | Section | Key Content |
|---|---|---|
| 1 | Cover + summary | File summary, lender A vs B side-by-side, quote expiration, contact |
| 2 | Why this rate | Waterfall chart: base rate -> adjustments -> final rate, per lender |
| 3 | Constraints | LTV / DSCR / term binding constraints, with 'what unbinds each' |
| 4 | Track A / Track B | Dual-track verdict, confidence score, deal-break curve |
| 5 | Scenarios | 5-toggle what-if table, with new PITIA / DSCR / Fit Tier per toggle |
| 6 | Assumptions appendix | Every input with provenance tag, source, timestamp, refresh date |

## Key Performance Indicators

| KPI | Baseline | 90-day | 365-day |
|---|---|---|---|
| Quote acceptance rate | ~35% | 55% | >70% |
| Borrower NPS | unmeasured | 30 | >50 |
| Time-to-trust (verdict to acceptance) | ~7 days | 4 days | <2 days |
| Quotes with full provenance tags | 0% | 60% | 100% |
| Operator overrides per quote | n/a | <3 | <1.5 |
| Borrower-initiated scenario toggles | 0 | ~3/quote | ~8/quote |

## Risks and Mitigations

The principal risk is information overload — a 6-page quote packet with 50 reason codes can overwhelm a borrower who is not financially sophisticated. Mitigation: the quote packet is layered — the cover page is one-glance, the comparison is two-minute, the assumptions appendix is deep-dive, and the borrower portal surfaces only the layer the borrower wants. A secondary risk is competitive intelligence leakage — a quote packet that decomposes the rate too thoroughly may help competitors reverse-engineer the lender's adjustment matrix. Mitigation: the adjustment matrix is shown to the borrower but the lender's name in the matrix is anonymized in any exportable format ('Lender A adjustment: +25bps at 75% LTV' rather than 'Griffin adjustment'). A tertiary risk is false precision — showing a DSCR of 1.05 implies more accuracy than the underlying rent comp (which may have ±$75/month uncertainty) supports. Mitigation: the confidence score is shown alongside every number, and a low-confidence number is flagged with an explicit 'verify before relying' marker.

CHAPTER 06

# Function 4 — Capital Partner Trust: Clean Files, Zero Defects

*Elite standard: capital partners trust your files because they are clean, complete, and low-defect. First-pass clean rate above 90%, defect rate below 2%, and every submission matches the target lender's exact stack — first time, every time.*

## Elite Standard

Capital partner trust is earned one file at a time, and lost one defect at a time. The elite operator sends files that the lender's underwriter can approve on first review — every required document is present, every number reconciles across documents (rent roll matches lease matches 1007 matches appraisal), every signature is in place, every compliance attestation is signed. The first-pass clean rate (the percentage of submitted files that the lender does not return for rework) exceeds 90%. The defect rate (defects per file, weighted by severity) is below 2%. The result is that the lender's underwriters move the operator's files to the top of the queue — they know that an operator's file is a 20-minute review, not a 2-hour treasure hunt.

This trust translates directly into turn time and pricing. A lender that trusts the operator's files will close in 18-21 days instead of 30-45 days, because the underwriter does not need to chase documents. A lender that trusts the operator's files will quote 12-25bps tighter on rate, because the lender's cost of carry and rework cost is lower. A lender that trusts the operator's files will grant the operator an exception when the operator needs one — a borderline DSCR, a borderline LTV, a borderline property type — because the lender knows the operator has earned the benefit of the doubt. Trust is the operator's most valuable capital asset, and it is built or destroyed by file quality.

| >90% | <2% | -40% | 100% |
|---|---|---|---|
| FIRST-PASS CLEAN RATE | DEFECT RATE PER FILE | TURN TIME VS PEER | STACK-MATCHED SUBMISSIONS |

## Current Gap

Today the platform has no file completeness engine, no document OCR, no defect scoring, and no submission package generator. The operator assembles the lender's required document stack manually (often from a checklist that is months out of date), emails the package to the lender, and waits for the lender's underwriter to identify the missing or inconsistent items. The first-pass clean rate is unmeasured but, based on operator feedback, likely below 50%. Defects are caught late — by the lender's underwriter, after the file has consumed underwriter time on both sides — which means each defect costs the operator credibility and the lender patience. There is no system that flags inconsistencies across documents before submission (rent roll shows $2,300/month, lease shows $2,150/month, 1007 shows $2,400/month — three documents, three numbers, no reconciliation).

## Required Upgrades

### 1. Pre-Submission Checklist Engine

Every lender's required document stack is encoded as a structured checklist in the lender guideline JSON. The stack varies by lender and by file shape — Griffin at 80% LTV requires a different stack than Kiavi at 70% LTV, and a cash-out refinance requires a different stack than a purchase. The pre-submission checklist engine takes a file shape and a target lender and returns the exact stack required: 1003 (with which sections), 1007 (with which comparables), 1008, rent roll (with which fields), lease (with which exhibits), title commitment (with which exceptions cleared), appraisal (with which approaches required), HOI declarations page, flood cert (if applicable), entity documents (operating agreement, EIN, articles, certificate of good standing), and compliance attestations. The operator cannot submit the file until every checkbox is green.

### 2. Document OCR and Cross-Validation

Every uploaded document is OCR'd and the key fields are extracted into a structured file schema. The cross-validation engine then checks for consistency across documents: does the rent on the rent roll match the rent on the lease (within $25)? Does the property address on the title commitment match the address on the appraisal (exact match)? Does the borrower name on the 1003 match the name on the operating agreement (exact match, with entity suffix handling)? Does the loan amount on the 1003 match the loan amount on the lender's application (exact match)? Every inconsistency is flagged with a severity (critical / major / minor) and an operator action (reconcile / document explanation / re-pull). The engine catches 80% of defects before submission.

### 3. Defect Scoring (File Heat Score)

Every file gets a heat score from 0 to 100, computed as a weighted sum of: completeness (40% — every required document present), consistency (30% — cross-document reconciliation clean), freshness (15% — every document within its validity window), and compliance (15% — every attestation signed, every disclosure delivered). A score above 85 is green (submit); 70-85 is amber (review with senior operator before submit); below 70 is red (do not submit — return to borrower for rework). The heat score is shown to the operator before submission, and the historical heat score is tracked per operator (an operator whose average heat score is 75 needs training; an operator whose average is 92 is a mentor).

### 4. Submission Package Generator

Once the heat score is green, the submission package generator produces the package in the exact format the target lender expects — bookmarks named to the lender's convention, files in the lender's preferred order, naming convention enforced, PDFs merged or separated as the lender requires, and a cover sheet with the file summary and lender-specific submission routing. The package is uploaded to the lender's portal (or emailed to the lender's submission address) directly from the platform, with a delivery confirmation logged. The generator eliminates the 30-60 minutes of manual packaging that currently precedes every submission.

### 5. Reconciliation Engine (Title vs Appraisal vs Survey)

The reconciliation engine specifically catches the high-cost mismatches that cause closing delays: title exception that the appraisal did not flag (encroachment, easement), survey discrepancy with the legal description on title, appraisal value that does not support the loan amount (LTV recompute), property condition on appraisal that contradicts the inspection report. These mismatches are the principal cause of closing-table surprises, and catching them pre-submission reduces closing delays by an estimated 40%. The engine produces a reconciliation report that goes into the file as an internal document, not as part of the lender submission.

### 6. Compliance Attestations

Every file carries a compliance attestation checklist: ECOA / Reg B notice delivered, fair-lending disclosure delivered, state-specific disclosure delivered (e.g., Texas 12-day letter, Minnesota private mortgage banker disclosure), NMLS identifier on every communication, borrower's right to receive appraisal copy delivered. The checklist is jurisdiction-aware (the compliance requirements differ by state and by property type) and lender-aware (some lenders require additional attestations). Attestations are signed electronically and logged with a tamper-evident timestamp. The compliance overlay exists to make every file defensible in the event of a borrower complaint or a regulatory audit.

## Defect Taxonomy

| Severity | Defect Class | Example | Cost if Caught Late |
|---|---|---|---|
| Critical | Identity mismatch | Borrower name on 1003 vs operating agreement | Closing delay 5-10 days |
| Critical | Title defect | Unreleased lien on title commitment | Closing delay 7-21 days |
| Critical | Value shortfall | Appraisal comes in $15K under contract | Re-trade or collapse |
| Major | Rent inconsistency | Rent roll $2,300 vs lease $2,150 (DSCR impact) | Re-underwrite, 3-5 day delay |
| Major | Missing entity doc | Certificate of good standing not in file | 1-3 day delay |
| Major | Insurance coverage gap | HOI dwelling limit below loan amount | 2-5 day delay |
| Minor | Stale document | Bank statements >60 days old | 1-2 day delay |
| Minor | Missing signature | Borrower unsigned on page 4 of 1003 | 1 day delay |
| Minor | Wrong format | Lender wants PDF, file submitted as JPEG | Same-day rework |

## New Code Modules

| Module | Purpose | Est. LoC |
|---|---|---|
| fileCompletenessEngine.ts | Per-lender stack checklist + green-light gate | ~1,200 |
| documentOcr.ts | OCR pipeline (Tesseract / cloud OCR), field extraction | ~1,800 |
| documentReconciler.ts | Cross-document consistency check, severity classifier | ~960 |
| defectScorer.ts | Heat score computation, per-operator tracking | ~430 |
| submissionPackGenerator.ts | Per-lender format compliance + upload / email | ~1,100 |
| titleAppraisalReconciler.ts | Title / appraisal / survey reconciliation | ~680 |
| complianceAttestations.ts | Jurisdiction-aware attestation checklist + e-sign | ~840 |
| lenderStackProfiles.ts | Structured per-lender stack JSON (25+ entries) | ~1,400 |

## Key Performance Indicators

| KPI | Baseline | 90-day | 365-day |
|---|---|---|---|
| First-pass clean rate | ~50% | 75% | >90% |
| Defect rate per file (weighted) | unmeasured | 5% | <2% |
| Median turn time (submission to clear) | 30-45 days | 25 days | 18-21 days |
| Files requiring lender rework | ~50% | 25% | <10% |
| Critical defects caught pre-submission | 0% | 60% | >90% |
| Average file heat score | n/a | 78 | >88 |

## Risks and Mitigations

The principal risk is OCR accuracy. OCR is reliable on clean scanned PDFs but degrades on photographs, fax-quality scans, and documents with handwriting. Mitigation: the OCR pipeline uses confidence scoring on every extracted field, and low-confidence extractions are routed to manual verification; the operator never sees a 'clean' heat score that is actually based on an OCR error. A secondary risk is checklist drift — lender stack requirements change over time, and a stale checklist produces confident wrong submissions. Mitigation: the checklist JSON carries a verification date and is updated on the same weekly cadence as the lender guideline diff engine; a lender whose checklist has not been verified in 90 days produces a warning in the completeness engine. A tertiary risk is over-engineering — the completeness engine could become so strict that it slows the operator down without materially improving quality. Mitigation: the engine has an operator override path with audit logging, and override frequency is reviewed monthly; overrides above 10% on a particular checklist item trigger a checklist review (the checklist is probably wrong, not the operator).

CHAPTER 07

# Function 5 — Distribution: Repeatable Referral Channels

*Elite standard: 60%+ of revenue from repeat referral channels, not random leads. The shop has named channels with economics, not a marketing budget with hopes.*

## Elite Standard

The elite DSCR shop does not buy leads; it cultivates channels. A channel is a repeatable source of files with a known unit economics: cost per submitted file, cost per closed file, average revenue per closed file, repeat interval (time from one closed loan to the next from the same source), and channel lifetime value. The elite shop tracks at least five active channels — agent referrals, repeat investors, wholesalers, CPA / financial advisor referrals, and loan officer syndicate — and no single channel accounts for more than 40% of revenue (concentration risk). Sixty percent of revenue comes from channels that produced a closed loan in the prior 12 months; the remaining 40% comes from new channel development, which is the pipeline that becomes next year's repeat channels.

Channel discipline is what separates a sustainable DSCR shop from a broker that lives or dies by the next Zillow lead. The shop that has repeat channels can plan staffing, can negotiate better lender terms (because the lender knows the shop's pipeline is predictable), and can survive a marketing-budget cut or a Google Ads suspension. The shop without repeat channels is one algorithm change away from extinction. The platform's job is to make channel cultivation mechanical: every channel has a portal, every file is attributed to a channel at intake, and channel economics are computed automatically so the operator can see which channels are paying off and which are wasting time.

| >60% | <180d | 5+ | <40% |
|---|---|---|---|
| REVENUE FROM REPEAT CHANNELS | REPEAT-DEAL INTERVAL | ACTIVE CHANNELS | SINGLE-CHANNEL CONCENTRATION |

## Current Gap

Today the platform has no distribution layer. Files arrive by phone, email, and word-of-mouth; there is no intake portal, no channel attribution, and no channel economics tracking. The operator cannot answer the question 'which of my referral sources produced closed loans last quarter?' with any confidence, because the data was never captured. There is no investor CRM, so the shop does not know when a repeat investor's equity buildup crosses the threshold that triggers a cash-out refinance opportunity. There is no agent referral portal, so agents who could be sending repeat business have no easy way to submit files and track their referrals' progress. The shop is operating as a transaction processor, not as a channel cultivator.

## Required Upgrades

### 1. Agent Referral Portal

Real estate agents who work with investors are the highest-quality file source in DSCR lending — they have a relationship with the borrower, they understand the transaction, and they have a financial incentive (their buyer's offer is stronger with a DSCR pre-approval in hand). The agent referral portal lets an agent submit a file in under 5 minutes (property address, borrower name, transaction basics), receive an instant pre-flight verdict (red/amber/green), and track the file's progress through underwriting to closing. The portal also shows the agent their referral history — files submitted, files closed, commission earned (if the shop pays referral fees), and average turn time. Agents who refer repeatedly become 'preferred partners' with priority underwriting and a direct line to a senior operator.

### 2. Wholesaler Intake API

Wholesalers move volume but produce highly variable file quality — a single wholesaler may submit 30 files a month, of which 5 are fundable. The wholesaler intake API receives files programmatically (JSON payload with property and borrower basics), runs the pre-flight gate automatically, and returns a verdict to the wholesaler within 90 seconds. Files that fail the pre-flight gate are auto-rejected with a reason code, so the wholesaler does not waste the operator's time on unfundable files. Files that pass the pre-flight gate are queued for operator review, with the wholesaler's submission history visible (this wholesaler's pass rate over the last 90 days is 22%, so the operator knows what to expect). The API has rate limits per wholesaler and a quality floor — wholesalers whose closed-loan rate falls below 5% over a rolling 90-day window have their API access suspended.

### 3. Repeat Investor Dashboard

The repeat investor dashboard is the platform's principal retention tool. Every investor who has closed a loan with the shop is tracked in the investor CRM, which monitors their portfolio (property count, total debt, total equity, weighted-average DSCR, weighted-average rate) and triggers a 'next-deal opportunity' alert when: (a) equity buildup crosses 25% (cash-out refinance opportunity); (b) a balloon or rate-reset is approaching within 12 months (refinance opportunity); (c) the investor's portfolio DSCR has improved such that they now qualify for better terms (re-quote opportunity); (d) the investor has not closed a loan in 18 months but their portfolio suggests they are active (re-engagement opportunity). The alert goes to the operator, who calls the investor with a specific opportunity, not a generic check-in.

### 4. Loan Officer Syndicate (Correspondent Channels)

Loan officers at banks and credit unions who cannot do DSCR loans themselves (their institution does not offer the product) are a high-quality referral source — they have a relationship with the borrower and a financial incentive to refer (revenue share or co-marketing). The syndicate program enrolls these LOs as correspondents, gives them a co-branded portal (the LO submits the file under their name, the shop does the underwriting, the LO gets a revenue share on close), and tracks each correspondent's volume, quality, and revenue. Top correspondents graduate to 'senior correspondent' status with a higher revenue share and priority underwriting. The syndicate program is the principal channel for scaling beyond the founder's personal network.

### 5. CPA / Financial Advisor Referral Program

CPAs and financial advisors who serve high-income clients (physicians, attorneys, business owners) are an under-tapped referral source for DSCR loans — their clients often want to diversify into real estate but do not have a relationship with a DSCR lender. The referral program enrolls CPAs and advisors, gives them a one-page pitch deck (the tax-aligned DSCR pitch: 'your client can use DSCR financing to acquire rental property without W2 income qualification, and the interest is deductible against the rental income'), and tracks referrals through closing.

The program is intentionally low-friction (CPAs do not want to learn underwriting) — the CPA's role is to make the introduction, and the shop takes it from there. Revenue share is paid only on closed loans.

### 6. Channel Attribution and Economics Tracking

Every file at intake is attributed to a channel (agent referral, repeat investor, wholesaler, correspondent, CPA, inbound). The attribution is recorded at the moment of intake, not retroactively. Channel economics are computed monthly: cost per submitted file (channel-specific costs — referral fees, marketing spend, API costs), cost per closed file, average revenue per closed file, gross margin per closed file, repeat interval (median time from one closed loan to the next from the same channel), and channel lifetime value. The economics are shown in a dashboard that lets the operator see at a glance which channels are profitable, which are break-even, and which are losing money. Channels that are losing money for two consecutive quarters are placed on probation; channels that are highly profitable get additional investment.

## Channel Economics Framework

| Channel | Cost per file | Cost per close | Avg revenue | Repeat interval | Target mix |
|---|---|---|---|---|---|
| Agent referrals | $80 | $800 | $7,500 | 120 days | 25% |
| Repeat investors | $50 | $200 | $8,200 | 180 days | 20% |
| Wholesalers | $30 | $450 | $6,800 | 90 days | 15% |
| Correspondent LOs | $20 | $150 | $7,000 | 150 days | 15% |
| CPA / advisors | $40 | $300 | $9,500 | 365 days | 10% |
| Inbound (website / SEO) | $120 | $1,800 | $7,200 | 270 days | 10% |
| New channel development | n/a | n/a | n/a | n/a | 5% |

## New Code Modules

| Module | Purpose | Est. LoC |
|---|---|---|
| agentPortal.ts | Agent-facing submission + status portal | ~2,200 |
| wholesalerIntake.ts | JSON API + pre-flight auto-reject + quality floor | ~1,400 |
| investorCRM.ts | Investor portfolio tracking + next-deal triggers | ~1,800 |
| correspondentSyndicate.ts | LO enrollment, co-branded portal, revenue share | ~1,600 |
| cpaAdvisorProgram.ts | CPA enrollment, pitch deck, referral tracking | ~720 |
| channelAttribution.ts | Channel tagging at intake + monthly economics | ~960 |
| channelDashboard.ts | Channel P&L; dashboard for operator | ~1,100 |

## Key Performance Indicators

| KPI | Baseline | 90-day | 365-day |
|---|---|---|---|
| % revenue from repeat channels | ~25% | 40% | >60% |
| Active channels (>1 close/quarter) | 2 | 4 | 5+ |
| Single-channel concentration | ~70% | <55% | <40% |
| Repeat-deal interval (median) | unmeasured | 240d | <180d |
| Cost per closed file (blended) | unmeasured | $750 | <$500 |
| Channel-attributed files (vs random) | ~30% | 70% | 100% |

## Risks and Mitigations

The principal risk is channel concentration — a single agent or wholesaler who produces 40% of revenue has leverage that can be used to demand better terms or to defect to a competitor. Mitigation: the platform flags any channel that crosses 30% concentration and triggers active development of alternative channels; no channel is allowed to exceed 40% without explicit operator acknowledgement and a diversification plan. A secondary risk is referral fee compliance — RESPA prohibits kickbacks for referrals on owner-occupied loans, and while DSCR is generally investment-property (not RESPA-covered), some states have broader prohibitions and some structures (revenue share on a loan that the shop underwrites but does not fund) can be characterized as fee splitting. Mitigation: all referral arrangements are reviewed by counsel before implementation, revenue share is paid only on closed loans as a consulting fee (not a referral fee), and the program is structured to comply with the most restrictive state's law. A tertiary risk is wholesaler file quality — wholesalers often submit files with inflated rent comps or hidden title issues that pass the pre-flight gate but fail in underwriting. Mitigation: the wholesaler quality floor (5% closed-loan rate over 90 days) auto-suspends low-quality wholesalers, and the pre-flight gate is tuned to be more aggressive on wholesaler-submitted files (lower confidence threshold for rent comps, mandatory title pull before verdict).

CHAPTER 08

# Function 6 — Risk Discipline: Decline Bad Files Early

*Elite standard: hard decline gates + soft warning gates, all logged, with ECOA-compliant adverse action. False-decline below 5%, and every declined file feeds the pattern library so the next bad file is caught faster.*

## Elite Standard

Risk discipline is the function most operators skip, because declining a file feels like leaving money on the table. The elite operator understands the opposite: every bad file that enters the pipeline consumes underwriter time, creates borrower ill-will when it inevitably declines late, exposes the shop to fair-lending scrutiny if the decline pattern is not defensible, and displaces a good file that could have been worked instead. The elite shop declines bad files at intake, not at underwriting — the pre-flight gate (Function 1) catches the obvious reds, and the decline gate (this function) catches the subtler ones. The false-decline rate (files declined that would have closed) is below 5%, and every decline produces a regulator-ready adverse action notice.

Risk discipline is also a feedback loop. Every decline is tagged with a reason code, and the reason codes are mined quarterly for patterns: 'we declined 14 condotels last quarter, all for the same lender-ineligibility reason — we should add this to the pre-flight gate so future condotels are declined at intake, not after underwriting time is spent.' This pattern library is the platform's institutional memory — it is what makes the shop smarter over time, not just bigger. A shop without a pattern library makes the same mistakes repeatedly; a shop with one becomes more efficient every quarter.

| <5% | >4 hrs | 100% | >60% |
|---|---|---|---|
| FALSE-DECLINE RATE | TIME SAVED PER DECLINED FILE | ADVERSE ACTION COMPLIANCE | DECLINES AT INTAKE (NOT UNDERWRITING) |

## Current Gap

Today the platform has no automated decline gates — every decline is an operator judgment, applied late in the workflow after underwriting time has been spent. There is no adverse action letter generator, which means declines are communicated verbally or by email without the regulatory disclosure that ECOA and Reg B require for credit applications. There is no pattern library, so the same bad file shapes recur quarter after quarter. The false-decline rate is unmeasured but, based on operator feedback, likely significant — operators decline files that would have closed because they are uncertain about a lender's effective minimum (Function 2 should fix this, but only if the decline-pattern learner is built).

## Required Upgrades

### 1. Auto-Decline Rules (Hard Gates)

The hard decline gates are non-negotiable auto-declines that fire at intake, before any underwriting time is spent. A file that hits any hard gate is declined with a reason code and an adverse action notice; the operator cannot override a hard gate without senior approval and a logged exception. The hard gates are: (a) PITIA greater than qualifying rent (Track A DSCR below 1.00 at any lender in the matrix, with no rent comp upside); (b) flood zone AE or VE without proof of HOI; (c) HOA STR ban when the file's income strategy is STR; (d) prohibited state (Minnesota for entity prepay); (e) prohibited state-lender combination (e.g., Washington ARM for lenders with ARM product); (f) LLC structure mismatch (e.g., borrower is a trust but lender does not fund trusts); (g) FICO below the lowest lender floor in the matrix; (h) loan amount below the lowest lender minimum or above the highest lender maximum; (i) property type not in any lender's matrix (e.g., commercial mixed-use above 4 units).

### 2. Risk Scorecard with Stoplight UI

Files that pass the hard gates receive a risk scorecard with a stoplight verdict: red (auto-decline — already caught by hard gates, but the scorecard is the second line of defense), amber (manual review — the file has a borderline dimension that requires operator judgment), green (proceed). The scorecard considers: DSCR margin (how close to the lender minimum), FICO margin (how close to the lender floor), LTV margin (how close to the lender cap), reserve adequacy (months of PITIA in reserves), property type complexity (SFR is green, condotel is amber, mixed-use is red without specialist review), borrower complexity (single-member LLC is green, multi-member LLC with foreign nationals is amber, trust is red without attorney review). The scorecard is shown to the operator before the file proceeds to underwriting; an amber file requires a senior operator sign-off before it can be submitted to a lender.

### 3. Adverse Action Letter Generator

Every decline produces an ECOA / Reg B compliant adverse action notice within 30 days of the application date. The notice includes: the specific reason(s) for the decline (from a standardized reason code library — not vague language like 'did not meet guidelines'), the borrower's right to know the reasons (already provided), the borrower's right to a copy of any appraisal or valuation used (if applicable), the borrower's right to dispute the accuracy of information in a consumer report (if a credit report was used), and the contact information for the credit reporting agency (if applicable). The notice is generated automatically from the decline reason code, sent to the borrower electronically (with paper mail as an option), and logged with a tamper-evident timestamp. The compliance overlay exists to make every decline defensible.

### 4. 'Why We Declined' Pattern Library

Every decline is tagged with: reason code, file shape at intake, file shape at decline (if different — what did underwriting discover that intake missed?), lender that declined (if applicable), lender's stated reason (if different from the platform's reason), operator who declined, and a free-text operator note. The pattern library is mined quarterly for: (a) recurring decline patterns that should be promoted to hard gates (e.g., 'we declined 14 files last quarter for condotel ineligibility — promote this to a hard gate'); (b) recurring intake misses (e.g., '12 files that passed the pre-flight gate were declined for HOA STR ban that the HOA database did not have — refresh the HOA database'); (c) lender decline patterns (Function 2's decline-pattern learner); (d) operator-specific patterns (an operator who declines 30% of files for DSCR may be using an outdated lender minimum).

### 5. Lender-Side Decline Pattern Learning

This is the same capability described in Chapter 4 (Function 2), but it is restated here because it is also a risk-discipline capability. Every lender decline is tagged with the lender's stated reason and recorded against the lender's effective guideline matrix. Over time, the platform learns each lender's effective minimums — the actual DSCR, FICO, and LTV that result in approval, not the published minimums. This learning feeds back into the fit scorer (Function 2) and the hard decline gates (this function) — a lender with an effective DSCR minimum of 1.10

(despite a published 1.00) means that files with DSCR 1.00-1.10 to that lender are auto-flagged amber and routed to alternative lenders. This is the platform's institutional memory.

### 6. ECOA / Reg B Compliance Overlay

Beyond the adverse action notice, the platform maintains a fair-lending audit trail on every decline: the file's protected-class characteristics are NOT recorded (which would itself be a fair-lending violation if used in the decision), but the decline reason, the operator who declined, and the comparable files (files with similar shape that were approved) are logged. Quarterly, the platform runs a comparative file analysis: are declines disproportionately concentrated in any geographic area, property type, or loan amount band? If a disparity is detected, the platform flags it for compliance review — the disparity may be legitimate (e.g., a particular market has more HOA STR bans), but it must be explainable. This overlay is the platform's defense against a fair-lending examination.

## Decline Reason Code Library (Subset)

| Code | Reason | Gate Type | Adverse Action Required |
|---|---|---|---|
| DSCR-001 | Track A DSCR below 1.00 at all lenders | Hard | Yes |
| FLOOD-001 | Flood zone AE/VE without HOI | Hard | Yes |
| HOA-001 | HOA prohibits STR (file income is STR) | Hard | Yes |
| STATE-001 | State prohibits entity prepay (MN) | Hard | Yes |
| LENDER-001 | State-lender combination prohibited (WA ARM) | Hard | Yes |
| ENTITY-001 | LLC structure not funded by any matrix lender | Hard | Yes |
| FICO-001 | FICO below lowest matrix lender floor | Hard | Yes (if credit pulled) |
| LOAN-001 | Loan amount outside matrix lender range | Hard | Yes |
| PROPTYPE-001 | Property type not in matrix (e.g., >4 units) | Hard | Yes |
| DSCR-002 | DSCR within 0.05 of lender minimum (borderline) | Soft | Only if declined |
| FICO-002 | FICO within 20 points of lender floor | Soft | Only if declined |
| LTV-001 | LTV within 2% of lender cap | Soft | Only if declined |
| RESERVE-001 | Reserves below 6 months PITIA | Soft | Only if declined |
| PROPTYPE-002 | Condotel — specialist lender required | Soft | Only if declined |
| ENTITY-002 | Multi-member LLC — attorney review required | Soft | Only if declined |

## New Code Modules

| Module | Purpose | Est. LoC |
|---|---|---|
| declineGate.ts | Hard gate evaluation + reason code assignment | ~620 |
| riskScorecard.ts | Stoplight computation + senior sign-off routing | ~540 |

| Module | Purpose | Est. LoC |
|---|---|---|
| adverseActionEngine.ts | ECOA/Reg B notice generation + e-delivery | ~880 |
| declinePatternLearner.ts | Quarterly pattern mining + hard-gate promotion | ~720 |
| reasonCodeLibrary.ts | Standardized reason code definitions (~60 codes) | ~440 |
| fairLendingAudit.ts | Comparative file analysis + disparity flagging | ~680 |
| exceptionOverride.ts | Senior approval workflow for hard-gate overrides | ~360 |

## Key Performance Indicators

| KPI | Baseline | 90-day | 365-day |
|---|---|---|---|
| False-decline rate | unmeasured | 12% | <5% |
| % declines at intake (not underwriting) | ~10% | 35% | >60% |
| Time saved per declined file | 0 hrs | 2 hrs | >4 hrs |
| Adverse action compliance (notices issued) | partial | 90% | 100% |
| Pattern library promotions (per quarter) | 0 | 3 | 8+ |
| Fair-lending audit pass rate | n/a | 100% | 100% |

## Risks and Mitigations

The principal risk is false-decline — a hard gate that declines a file which would have closed. False-declines are the most expensive error the platform can make, because they cost revenue directly and damage borrower trust. Mitigation: every hard gate has an override path with senior approval and audit logging; the false-decline rate is tracked monthly and any gate that exceeds 2% false-decline is recalibrated; the gate's threshold is initially conservative (decline only when the file is clearly unfundable) and tightens over time as the pattern library confirms the gate's accuracy. A secondary risk is fair-lending exposure from the comparative file analysis itself — if the analysis reveals a disparity and the shop does not act on it, the analysis becomes evidence in a fair-lending claim. Mitigation: the analysis is conducted under attorney-client privilege where possible, and any detected disparity triggers a documented investigation and remediation plan. A tertiary risk is decline-pattern library overfitting — a pattern observed in a small sample (e.g., 5 declines) may not generalize. Mitigation: pattern library promotions require a minimum sample size (10 instances) and a documented rationale; promotions are reviewed quarterly and can be reversed if subsequent data contradicts the pattern.

CHAPTER 09

# Cross-Cutting Capabilities

*The Six Functions share infrastructure. This chapter covers the four cross-cutting capabilities that span multiple functions: the data and provenance pipeline, the regulatory engine, the command-center UI, and the infrastructure that holds it all up.*

Each of the Six Functions has its own module inventory, but four capabilities are shared across functions and must be designed as such. Building them per-function would produce duplication, inconsistency, and maintenance debt; building them as cross-cutting capabilities produces leverage — one investment, multiple functions served. The four are: (1) the data and provenance pipeline that feeds Function 1, 2, 3, and 6; (2) the regulatory engine that gates Function 1, 4, and 6; (3) the command-center UI that operators, borrowers, and capital partners use to interact with all six functions; (4) the infrastructure — audit trail, role-based access, backup, performance budgets — that every function depends on.

## 1. Data and Provenance Pipeline

Every input to the platform — a rent comp, an insurance estimate, a tax figure, a lender guideline, a flood zone determination, an HOA rule — carries a provenance tag: [VERIFIED-Primary] (from the source of record, e.g., a county assessor API for taxes), [VERIFIED-Secondary] (from a credible secondary source, e.g., RentCast for rent comps), or [UNVERIFIED] (operator-entered or from an untrusted source). The provenance tag determines the input's weight in the confidence score (Function 1) and the quote packet's audit trail (Function 3). Inputs without provenance are forbidden — the platform refuses to use them.

The pipeline also manages data freshness. Every input carries a refresh cadence (county assessor data is annual, lender guidelines are weekly, rent comps are 30-day, flood zone determinations are 90-day). Inputs past their refresh cadence are flagged in the confidence score with a freshness penalty, and the operator is notified to re-pull. The pipeline also tracks stale-data alerts at the file level — if a file has been in the pipeline longer than 30 days, every input is re-checked for freshness before the file is submitted to a lender.

The verifiable source registry is the master catalog of every data source the platform uses: API endpoint, authentication method, refresh cadence, cost per pull, monthly cost ceiling, known accuracy gaps, and the function(s) that consume the source. The registry is the platform's data governance — a source not in the registry cannot be used, and adding a source to the registry requires documentation of accuracy, cost, and refresh cadence. This is what prevents the platform from accumulating untraceable inputs that produce confident wrong answers.

## 2. Regulatory Engine (50-State)

The regulatory engine encodes the legal constraints that affect every file: prepayment penalty law by state, STR regulations by jurisdiction, usury and rate caps by state, NMLS licensing requirements by state, and disclosure requirements by state and property type. The engine is the platform's compliance nervous system — it gates the pre-flight check (Function 1), the file completeness engine (Function 4), and the decline gates (Function 6). Every file is run through the regulatory engine at intake, and any regulatory red flag triggers either a hard decline (Function 6) or an amber flag for compliance review.

The prepayment penalty law sub-engine is the most complex component. It encodes the per-state rules: Minnesota prohibits entity prepay entirely; New Jersey prohibits prepay penalties for individual borrowers but permits them for entities; Illinois restricts prepay penalties to the first three years; Ohio permits a $116,356 (2026 indexed) penalty; Pennsylvania permits a $329,411 (2026 indexed) penalty; Mississippi uses the 5-4-3-2-1 step-down; Washington bans ARM products with prepay penalties entirely. Every loan quote is checked against the borrower's state and entity type, and the quote packet (Function 3) shows the applicable PPP rule with the citation. The 2026 indexed thresholds for Ohio and Pennsylvania are recomputed annually based on the published CPI adjustment.

The STR regulation sub-engine is the second most complex component. It encodes STR legality by jurisdiction (city + county + HOA layer, where HOA rules can be more restrictive than the city). The four-tier verdict — CLEAR, RESTRICTED, UNCERTAIN, PROHIBITED — gates whether STR income can be used in the Track A qualification. RESTRICTED means STR income is usable but with a discount or a cap; UNCERTAIN means the file is held for manual verification; PROHIBITED means STR income is not usable and the file must qualify on long-term rent. The STR database is refreshed weekly via a scrape of city ordinances and a partnership with a STR regulation data provider; the HOA database is refreshed quarterly via a manual review of HOA documents on file.

## 3. UI / UX — The Command Center

The command center is the operator's daily workspace. It has four primary surfaces, each tuned to a primary user. **The operator dashboard (single-deal view)** is the ten-minute verdict workflow — pre-flight gate, rent comp aggregator, lender auto-router, Track A/B verdicts, quote packet generation, decline gate. The dashboard is keyboard-driven (operators do not want to mouse) and surfaces the confidence score and the binding constraint at the top, so the operator can decide in 30 seconds whether to proceed or to triage. **The portfolio dashboard** is for the operator managing multiple files — pipeline view by stage (intake, pre-flight, underwriting, submission, clear-to-close, closed), with color-coded file health and exception alerts. **The lender intel console** is for the operator who maintains the lender matrix — recent guideline changes, relationship tier movements, decline-pattern summaries, and a 'what changed this week' digest.

**The channel economics dashboard** is for the operator who manages distribution — per-channel P&L;, concentration alerts, repeat-deal trigger alerts from the investor CRM, and wholesaler quality floor breaches. **The borrower portal** is the borrower's view of the quote packet — the 6-page PDF with interactive scenario toggles, the file status tracker, and the document upload interface. **The capital partner portal** is the lender's view of the submitted file — the document stack with bookmarks, the file summary, the operator's contact info, and the audit trail of every assumption. Each portal is role-based and shows only the information the role is entitled to see.

## 4. Infrastructure and Operations

The platform runs on Next.js 16 with TypeScript, Tailwind 4, and shadcn-ui on the frontend, with API routes and Prisma for persistence. The infrastructure layer that supports it has four components. **Immutable audit trail**: every assumption, every override, every lender submission, every decline, every adverse action notice is written to an append-only log with a tamper-evident hash chain. The log is the platform's defense in any subsequent dispute — it can prove what was assumed, what was overridden, what was disclosed, and when. **Role-based access control**: operators see their own files; senior operators see all files in their team; compliance officers see the audit trail but not the file contents; capital partners see only the files submitted to them; borrowers see only their own files. Access is logged.

**Backup and disaster recovery**: the database is backed up nightly with 30-day retention, the audit trail is replicated to a separate jurisdiction (for disaster recovery and for tamper-evidence — an attacker who compromises the primary cannot tamper with the replica without detection), and a documented runbook covers the principal failure modes (database loss, API outage at a critical data source, lender portal outage during submission). **Performance budgets**: the ten-minute verdict SLA (Function 1) is enforced as a per-step budget — pre-flight gate under 90 seconds, rent comp aggregator under 60 seconds, lender auto-router under 30 seconds, Track A/B computation under 30 seconds, quote packet generation under 60 seconds, with 30 seconds of operator judgment time reserved. A step that exceeds its budget triggers an alert and a performance investigation.

CHAPTER 10

# Implementation Roadmap — 4-Phase, 365-Day Plan

*Twelve months. Four phases. Phase 1 fixes the core. Phase 2 expands the matrix. Phase 3 builds distribution. Phase 4 completes the doctrine.*

The roadmap is sequenced so that each phase unlocks the next. Phase 1 (Days 0-30) fixes the broken core and adds the pre-flight gate — without a correct engine, every downstream investment amplifies the brokenness. Phase 2 (Days 31-90) expands the lender matrix and builds the file completeness engine — without lender depth and clean files, the distribution layer has nothing to distribute. Phase 3 (Days 91-180) builds the distribution layer and the risk discipline automation — without these, the shop cannot scale beyond the founder's personal network, and bad files continue to consume underwriter time. Phase 4 (Days 181-365) completes the regulatory engine, adds the Monte Carlo overlay, and reaches full GODMODE operating status. The phasing is aggressive but achievable with a focused team; it is also the minimum pace required to reach elite status before the next market reset.

## Phase 1 — Core Engine Fixes and Pre-Flight Gate (Days 0-30)

**Deliverables:** (1) Track A / Track B engine correctness audit, with golden test suite locking the payment factor (8.25% → 0.0075127, 7.00% → 0.006653), Track A DSCR (1.05 at 7.00%, 0.96 at 8.25%), rent breakeven (4.9%), and deal-break rate (7.67% for the flagship transaction); (2) pre-flight gate MVP covering 4 of 7 dimensions (STR legality, zoning, flood, property type) with red/amber/green output; (3) rent comp aggregator MVP covering RentCast and Rentometer (AirDNA in Phase 2); (4) state PPP engine operational with annual indexing logic for OH and PA; (5) confidence score v1 with source timeliness and source quantity components.

**Exit criteria:** golden test suite passes 100% on every commit; pre-flight gate catches 80% of files that should be auto-declined (validated against a 50-file retrospective sample); rent comp aggregator reduces manual rent lookup time from 8 minutes to 90 seconds; state PPP engine correctly identifies prohibited-state files with zero false-positives on the retrospective sample. **Dependencies:** RentCast and Rentometer API contracts signed; flood zone API selected and contracted; STR regulation database licensed. **Risk register:** API contract delay (mitigation: 2-week buffer; fallback to manual rent comp for first 30 days); STR database licensing cost overruns (mitigation: tiered pricing negotiation; start with top-50 MSAs only).

## Phase 2 — Lender Expansion and File Quality (Days 31-90)

**Deliverables:** (1) lender matrix expanded from 7 to 14 verified lenders (Visio, LendingOne, RCN, Anchor, Velocity, Park Place, plus Lima One STR and Griffin STR verification); (2) guideline diff engine operational with weekly cadence and pipeline impact analysis; (3) fit scorer v1 with Fit Tier mapping per lender; (4) two-quote rule enforced in quote packet generator; (5) file completeness engine MVP for top 5 lenders (Griffin, Kiavi, Lima One, Visio, LendingOne); (6) document OCR pipeline operational for 1003, 1007, rent roll, lease, title, and HOI; (7) defect scorer v1 with heat score; (8) submission package generator for top 5 lenders; (9) 'Why this rate' explainer MVP; (10) PDF quote packet v1 (4 pages).

**Exit criteria:** 14 lenders verified and live in fit scorer; guideline diff catches 100% of material changes within 7 days; two-quote rule enforced on 100% of quotes; file completeness engine reduces first-pass defect rate by 50% on the top 5 lenders; defect heat score averages above 80 on submitted files; quote packet delivered within 30 minutes of verdict on 90% of files. **Dependencies:** Phase 1 exit criteria met; OCR pipeline vendor selected; lender portal API access for 5 priority lenders. **Risk register:** lender verification slower than expected (mitigation: parallel verification tracks, 2 lenders per operator-week); OCR accuracy below 90% (mitigation: confidence-scored extraction with manual verification fallback).

## Phase 3 — Distribution and Risk Discipline (Days 91-180)

**Deliverables:** (1) lender matrix expanded to 20 verified lenders; (2) relationship tracker operational with quarterly tier computation; (3) decline pattern learner operational with quarterly pattern mining; (4) agent referral portal MVP; (5) wholesaler intake API MVP with pre-flight auto-reject; (6) investor CRM MVP with next-deal triggers (equity buildup, balloon approaching, DSCR improvement); (7) channel attribution and economics dashboard; (8) hard decline gates operational for 9 reason codes; (9) risk scorecard with stoplight UI; (10) adverse action letter generator for 30 reason codes; (11) fair-lending audit overlay with quarterly comparative file analysis; (12) tax and insurance estimator integrations live.

**Exit criteria:** 20 lenders verified; agent referral portal producing 5+ submitted files per week; investor CRM triggering 3+ next-deal opportunities per month; channel economics dashboard showing per-channel P&L; for all active channels; hard decline gates catching 60% of declined files at intake (vs 10% baseline); adverse action compliance 100% on declined files. **Dependencies:** Phase 2 exit criteria met; channel partner agreements in place for top 3 agents and top 2 wholesalers; ECOA / Reg B compliance review of adverse action templates by counsel. **Risk register:** agent portal adoption slower than expected (mitigation: in-person onboarding for top 10 agents; referral fee pilot program); fair-lending analysis reveals a disparity (mitigation: pre-planned investigation protocol with counsel; documented remediation framework).

## Phase 4 — GODMODE: Regulatory Engine and Monte Carlo (Days 181-365)

**Deliverables:** (1) lender matrix at 25+ verified lenders; (2) regulatory engine complete (50-state PPP, STR regulations for top-100 MSAs, usury / rate caps, NMLS licensing, state-specific disclosures); (3) Monte Carlo survivorship overlay operational with 10,000-trial Track B output; (4) deal-break curve solver operational across LTV / rent / DSCR sweep; (5) live pricing grid integration for top 10 lenders (Optimal Blue / PPE / lender-direct); (6) correspondent syndicate program live with 5+ enrolled LOs; (7) CPA / advisor referral program live with 10+ enrolled advisors; (8) constraint visualizer with 'what unbinds this' interactive; (9) scenario toggles live in borrower portal; (10) full audit trail with role-based access; (11) command center UI v2 with operator / portfolio / lender / channel dashboards.

**Exit criteria (GODMODE status):** 95% of verdicts under 10 minutes; 90% confidence calibration; fit accuracy above 85% (will-close prediction); first-pass clean rate above 90%; defect rate below 2%; 60%+ of revenue from repeat channels; false-decline below 5%; 100% adverse action compliance; full regulatory engine covering 50 states; Monte Carlo overlay on every Track B verdict; live pricing on top 10 lenders. **Dependencies:** Phase 3 exit criteria met; Optimal Blue or equivalent pricing API contracted; full-time compliance officer in seat; 4-person operator team trained on the platform. **Risk register:** regulatory engine scope creep (mitigation: ship 50-state PPP first, then STR for top-50 MSAs, then long-tail STR; do not block GODMODE status on long-tail regulatory coverage); Monte Carlo runtime exceeds budget (mitigation: pre-compute survivorship distributions per file shape, cache results).

## Roadmap Summary (Gantt-Style)

| Workstream | Days 0-30 | Days 31-90 | Days 91-180 | Days 181-365 |
|---|---|---|---|---|
| Core engine correctness | ■ P1 | | | |
| Pre-flight gate | ■ P1 | expand | expand | complete |
| Rent comp aggregator | MVP P1 | expand | AirDNA | complete |
| Lender matrix | audit | → 14 | → 20 | → 25+ |
| Guideline diff engine | | ■ P2 | expand | complete |
| Fit scorer + relationship | | ■ P2 | ■ P3 | complete |
| File completeness engine | | ■ P2 | expand | complete |
| OCR + reconciler | | ■ P2 | expand | complete |
| Quote packet + explainer | | ■ P2 | expand | complete |
| Agent / wholesaler portal | | | ■ P3 | expand |
| Investor CRM | | | ■ P3 | expand |
| Channel attribution | | | ■ P3 | complete |
| Decline gates + adverse action | | | ■ P3 | complete |
| Regulatory engine (50-state) | PPP | PPP expand | PPP + STR top-50 | complete 50-state |
| Monte Carlo overlay | | | | ■ P4 |
| Live pricing grid | | | pilot | top 10 lenders |

CHAPTER 11

# Success Metrics and KPIs

*What gets measured gets managed. The KPI framework maps each of the Six Functions to leading and lagging indicators, with target values and measurement cadence.*

The KPI framework is the platform's scorecard. It distinguishes leading indicators (which predict future performance — time-to-verdict, first-pass clean rate, false-decline rate) from lagging indicators (which confirm past performance — closed-loan volume, revenue, margin). Leading indicators are measured weekly and reviewed monthly; lagging indicators are measured monthly and reviewed quarterly. The framework is designed so that an operator can tell, from the leading indicators alone, whether the doctrine is being lived — without waiting for the revenue to confirm or deny it.

The north-star metric is closed-loan velocity per operator per month — the number of closed loans a single operator can produce, normalized for file complexity. This metric captures all six functions: a high velocity means the operator is fast (Function 1), accurate (Function 2), trusted (Function 3), clean (Function 4), channeled (Function 5), and disciplined (Function 6). The target is 12 closed loans per operator per month at steady state (Phase 4), up from a baseline of 4-6 today. Every other KPI in the framework either contributes to or measures a component of this north-star.

## Master KPI Dashboard

| Function | KPI | Type | Cadence | Target (365d) |
|---|---|---|---|---|
| 1 Scenario | Median time-to-verdict | Leading | Weekly | <10 min |
| 1 Scenario | % verdicts under 10 min | Leading | Weekly | 95% |
| 1 Scenario | Confidence calibration (Brier score) | Leading | Monthly | <0.10 |
| 1 Scenario | False GO rate | Lagging | Monthly | <5% |
| 2 Guidelines | Verified lenders in matrix | Leading | Monthly | 25+ |
| 2 Guidelines | Two-quote compliance | Leading | Weekly | 100% |
| 2 Guidelines | Fit accuracy (will-close prediction) | Lagging | Quarterly | >85% |
| 2 Guidelines | Guideline diff latency | Leading | Weekly | <7 days |
| 3 Borrower Trust | Quote acceptance rate | Lagging | Monthly | >70% |
| 3 Borrower Trust | Borrower NPS | Lagging | Quarterly | >50 |
| 3 Borrower Trust | Quotes with provenance tags | Leading | Weekly | 100% |
| 4 Partner Trust | First-pass clean rate | Leading | Weekly | >90% |
| 4 Partner Trust | Defect rate per file | Leading | Weekly | <2% |
| 4 Partner Trust | Median turn time (submit to clear) | Lagging | Monthly | 18-21 days |

| Function | KPI | Type | Cadence | Target (365d) |
|---|---|---|---|---|
| 4 Partner Trust | Critical defects caught pre-submission | Leading | Monthly | >90% |
| 5 Distribution | % revenue from repeat channels | Lagging | Monthly | >60% |
| 5 Distribution | Active channels | Leading | Monthly | 5+ |
| 5 Distribution | Single-channel concentration | Leading | Monthly | <40% |
| 5 Distribution | Repeat-deal interval (median) | Lagging | Quarterly | <180 days |
| 6 Risk Discipline | False-decline rate | Lagging | Monthly | <5% |
| 6 Risk Discipline | % declines at intake | Leading | Weekly | >60% |
| 6 Risk Discipline | Adverse action compliance | Leading | Weekly | 100% |
| 6 Risk Discipline | Fair-lending audit pass rate | Lagging | Quarterly | 100% |
| NORTH STAR | Closed loans / operator / month | Lagging | Monthly | 12 |

## Calibration Discipline

Calibration is harder than accuracy, and it is the metric that most operators fail to track. An uncalibrated operator who is confident 90% of the time but correct 75% of the time is more dangerous than a calibrated operator who is confident 75% of the time and correct 75% of the time — the uncalibrated operator produces false certainty that leads to wasted underwriting time and damaged borrower trust. The platform tracks calibration via a Brier score on every confidence-weighted verdict: a verdict of '90% confident this file will close' that does close contributes 0.01 to the Brier score; a verdict that does not close contributes 0.81. The Brier score is averaged across all verdicts per operator per month; a Brier score below 0.10 is elite, 0.10-0.18 is acceptable, above 0.18 requires recalibration training.

Calibration is also tracked at the platform level — the platform's confidence scores must themselves be calibrated. If the platform says '90% confidence' on a verdict and only 75% of those verdicts are correct, the platform's confidence model is mis-calibrated and must be retrained. The platform Brier score is computed monthly across all verdicts and is the principal measure of the platform's judgment quality. A platform Brier score below 0.10 is the GODMODE standard; above 0.15 is unacceptable.

## Anti-KPIs (What Not to Measure)

**METRICS THAT DISTRACT, NOT DRIVE**

**Quote volume** — measuring the number of quotes sent rewards speed over accuracy and produces false-positive quotes that erode trust. **Lender count** — measuring the raw number of lenders in the matrix rewards breadth over depth; a matrix with 25 verified lenders and 5 A-tier relationships beats a matrix with 40 unverified lenders and 0 relationships. **Marketing impressions** — measuring ad reach rewards spend over channel fit; the doctrine is repeat channels, not random reach. **Underwriting hours** — measuring hours worked rewards effort over judgment; the doctrine is faster verdicts, not more hours. **Files in pipeline** — measuring pipeline size rewards intake over discipline; the doctrine is declining bad files at intake, not stockpiling them. Every KPI that is not in the master dashboard above is, by definition, a distraction.

CHAPTER 12

# Appendices

*Reference material: lender verification registry, state PPP matrix, data source catalog, and code module inventory.*

## Appendix A — Target Lender Verification Registry (Year 1)

The lender registry is the master list of every lender in the matrix, with verification status, key guideline parameters, and relationship tier. The registry is the source of truth for Function 2 (Guideline Intelligence) and Function 4 (Capital Partner Trust). Lenders marked VERIFIED have primary-sourced guideline data; those marked TARGET are slated for verification in the indicated quarter; those marked UNVERIFIED have secondary-sourced data and require primary verification before they can be used in the fit scorer.

| Lender | LTV | DSCR min | FICO min | Max loan | STR? | Status |
|---|---|---|---|---|---|---|
| Griffin | 75-85% | 1.00 | 660 | $3M | Yes | VERIFIED |
| Defy | 80% | 1.00 | 660 | $2.5M | Yes | VERIFIED |
| Easy Street | 82% | 1.00 | 660 | $3M | Yes | VERIFIED |
| Lima One | 76% | 1.00 | 660 | $2M | Yes | VERIFIED |
| New Silver | 72% | 1.00 | 660 | $2M | Yes | VERIFIED |
| Kiavi | 70% | 1.00 | 660 | $3M | Yes | VERIFIED |
| Deephaven | 65% | 1.00 | 660 | $2M | Yes | VERIFIED |
| Visio Lending | 75-80% | 1.00 | 680 | $2M | Yes | TARGET — Q2 |
| LendingOne | 70-80% | 1.00 | 660 | $2M | Yes | TARGET — Q2 |
| RCN Capital | 65-75% | 1.00 | 660 | $5M | Limited | TARGET — Q2 |
| Anchor Loans | 70-80% | n/a | 660 | $5M | No | TARGET — Q3 |
| Velocity Mortgage | 65-80% | 1.00 | 660 | $1.5M | Yes | TARGET — Q3 |
| Park Place Finance | 70-80% | 1.00 | 660 | $2M | Yes | TARGET — Q3 |
| Anchor Six Lending | 70-80% | 1.00 | 660 | $2M | Yes | TARGET — Q4 |
| Broadmark Realty | 65-75% | n/a | 660 | $5M | No | TARGET — Q4 |
| CoreVest Finance | 70-80% | 1.00 | 680 | $50M | Yes | TARGET — Q4 |
| Marquee Funding | 65-75% | 1.00 | 660 | $5M | No | TARGET — Q4 |
| Pacwest Capital | 70-80% | 1.00 | 660 | $5M | Yes | TARGET — Q4 |
| Grand Coast Financial | 65-75% | 1.00 | 660 | $5M | No | TARGET — Q4 |
| Pinnacle DSCR | 70-80% | 1.00 | 660 | $2M | Yes | TARGET — Q4 |

| Lender | LTV | DSCR min | FICO min | Max loan | STR? | Status |
|---|---|---|---|---|---|---|
| Stratton Equities | 70-80% | 1.00 | 660 | $3M | Limited | TARGET — Q4 |
| Centerpoint Lending | 70-80% | 1.00 | 660 | $2M | Yes | TARGET — Q4 |

## Appendix B — State PPP Law Matrix (Selected Jurisdictions)

The state PPP matrix encodes the prepayment penalty law by state and borrower entity type. The matrix is the source of truth for Function 1 (pre-flight gate), Function 4 (compliance attestation), and Function 6 (decline gate). Entries marked VERIFIED have primary-sourced statutory citation; UNVERIFIED entries require confirmation before use in production. The 2026 indexed thresholds for Ohio and Pennsylvania are recomputed annually based on the published CPI adjustment; the values shown are for the 2026 calendar year.

| State | Entity Rule | Penalty Structure | 2026 Cap | Status |
|---|---|---|---|---|
| Minnesota | Entity prohibited | No PPP for entity borrowers | n/a | VERIFIED |
| New Jersey | Individual prohibited | PPP permitted for entities only | n/a | VERIFIED |
| Illinois | Restricted (3-year) | PPP permitted in first 3 years only | n/a | VERIFIED |
| Ohio | Permitted with cap | PPP permitted up to indexed threshold | $116,356 | VERIFIED (2026 indexed) |
| Pennsylvania | Permitted with cap | PPP permitted up to indexed threshold | $329,411 | VERIFIED (2026 indexed) |
| Mississippi | 5-4-3-2-1 step-down | PPP declines 5%/4%/3%/2%/1% by year | n/a | VERIFIED |
| Washington | ARM ban | No PPP on ARM products | n/a | UNVERIFIED — confirm |
| California | Permitted (regulated) | PPP permitted with disclosure | n/a | VERIFIED |
| Texas | Permitted (12-day letter) | PPP permitted with 12-day disclosure | n/a | VERIFIED |
| Florida | Permitted | PPP permitted, market-standard | n/a | VERIFIED |
| Arizona | Permitted | PPP permitted, market-standard | n/a | VERIFIED |
| Georgia | Permitted | PPP permitted, market-standard | n/a | VERIFIED |
| North Carolina | Permitted | PPP permitted, market-standard | n/a | VERIFIED |
| Tennessee | Permitted | PPP permitted, market-standard | n/a | VERIFIED |
| Colorado | Permitted (regulated) | PPP permitted with disclosure | n/a | VERIFIED |
| Nevada | Permitted | PPP permitted, market-standard | n/a | VERIFIED |
| Oregon | Restricted | PPP permitted with limits; verify | n/a | UNVERIFIED — confirm |
| Massachusetts | Restricted | PPP permitted with limits; verify | n/a | UNVERIFIED — confirm |

## Appendix C — Data Source Catalog

The data source catalog is the verifiable source registry — the master list of every external data source the platform uses, with API endpoint, refresh cadence, cost per pull, monthly cost ceiling, known accuracy gaps, and the consuming function(s). Sources not in the catalog cannot be used in production; adding a source requires documentation of accuracy, cost, and refresh cadence, plus approval by the operator desk.

| Source | Used For | Refresh | Cost/Pull | Functions |
|---|---|---|---|---|
| RentCast API | Long-term market rent | 30 days | $0.10 | F1, F3 |
| Rentometer API | Long-term rent percentile range | 30 days | $0.08 | F1, F3 |
| AirDNA API | STR projected revenue | 14 days | $0.50 | F1, F3 |
| MLS 1007 (manual) | Market rent opinion | Appraisal | n/a | F1, F3 |
| County assessor APIs | Property tax estimate | Annual | Free | F1, F3 |
| Millage database | Tax estimate fallback | Annual | Free | F1, F3 |
| FEMA flood service | Flood zone determination | 90 days | Free | F1, F6 |
| HOI estimation service | Insurance estimate | 30 days | $0.25 | F1, F3 |
| STR regulation DB | STR legality (city/county) | Weekly | License | F1, F6 |
| HOA document DB | HOA STR restrictions | Quarterly | Manual | F1, F6 |
| Optimal Blue / PPE | Live lender pricing | Daily | License | F2, F3 |
| Lender portal APIs | Lender guideline + status | Weekly | Free | F2, F4 |
| Tesseract / cloud OCR | Document extraction | Per upload | $0.01/pg | F4 |
| NMLS registry | Lender licensing verification | Monthly | Free | F4, F6 |
| State statute scrape | PPP / usury / disclosure | Quarterly | Free | F6, F9 |

## Appendix D — Code Module Inventory

The code module inventory is the master list of every TypeScript module to be built or modified, with the function(s) it serves, current status, and dependencies. Modules marked EXIST need correctness audit; NEW modules are to be built; UPGRADE modules need functional expansion. Total estimated lines of code across all modules: approximately 38,000. Total estimated build effort: 4-5 engineer-quarters, sequenced across the 4-phase roadmap.

| Module | Function | Status | Est. LoC |
|---|---|---|---|
| types.ts | Core | EXIST — audit | ~1,200 |
| engine.ts | F1 | EXIST — audit | ~2,000 |
| lenders.ts | F2 | EXIST — expand | ~1,800 |
| reserveEngine.ts | F1 | EXIST — expand | ~1,400 |
| sensitivity.ts | F1 | EXIST — upgrade | ~1,000 |

| Module | Function | Status | Est. LoC |
|---|---|---|---|
| loanOptimizer.ts | F1 | EXIST — upgrade | ~1,300 |
| strUnderwriting.ts | F1 | EXIST — audit | ~900 |
| preflightGate.ts | F1 | NEW | ~650 |
| rentCompAggregator.ts | F1 | NEW | ~820 |
| insuranceEstimator.ts | F1 | NEW | ~430 |
| taxEstimator.ts | F1 | NEW | ~380 |
| confidenceScorer.ts | F1 | NEW | ~290 |
| dealBreakCurve.ts | F1 | NEW | ~340 |
| monteCarloSurvivorship.ts | F1 | NEW | ~720 |
| lenderGuidelines.ts | F2 | NEW | ~2,400 |
| guidelineDiff.ts | F2 | NEW | ~640 |
| fitScorer.ts | F2 | NEW | ~520 |
| pricingGridFeed.ts | F2 | NEW | ~880 |
| relationshipTracker.ts | F2 | NEW | ~430 |
| declinePatternLearner.ts | F2 / F6 | NEW | ~570 |
| twoQuoteEnforcer.ts | F2 | NEW | ~210 |
| quoteExplainer.ts | F3 | NEW | ~680 |
| rateAdjustmentEngine.ts | F3 | NEW | ~540 |
| constraintVisualizer.ts | F3 | NEW | ~470 |
| scenarioToggles.ts | F3 | NEW | ~390 |
| pdfQuotePack.ts | F3 | NEW | ~1,100 |
| reasonCodes.ts | F3 | NEW | ~320 |
| auditLog.ts | F3 / Cross | NEW | ~430 |
| borrowerPortal.ts | F3 | NEW | ~1,800 |
| fileCompletenessEngine.ts | F4 | NEW | ~1,200 |
| documentOcr.ts | F4 | NEW | ~1,800 |
| documentReconciler.ts | F4 | NEW | ~960 |
| defectScorer.ts | F4 | NEW | ~430 |
| submissionPackGenerator.ts | F4 | NEW | ~1,100 |
| titleAppraisalReconciler.ts | F4 | NEW | ~680 |
| complianceAttestations.ts | F4 / Cross | NEW | ~840 |

| Module | Function | Status | Est. LoC |
|---|---|---|---|
| lenderStackProfiles.ts | F4 | NEW | ~1,400 |
| agentPortal.ts | F5 | NEW | ~2,200 |
| wholesalerIntake.ts | F5 | NEW | ~1,400 |
| investorCRM.ts | F5 | NEW | ~1,800 |
| correspondentSyndicate.ts | F5 | NEW | ~1,600 |
| cpaAdvisorProgram.ts | F5 | NEW | ~720 |
| channelAttribution.ts | F5 | NEW | ~960 |
| channelDashboard.ts | F5 | NEW | ~1,100 |
| declineGate.ts | F6 | NEW | ~620 |
| riskScorecard.ts | F6 | NEW | ~540 |
| adverseActionEngine.ts | F6 | NEW | ~880 |
| reasonCodeLibrary.ts | F6 | NEW | ~440 |
| fairLendingAudit.ts | F6 / Cross | NEW | ~680 |
| exceptionOverride.ts | F6 | NEW | ~360 |
| regulatoryEngine.ts | Cross | NEW | ~3,200 |
| statePppMatrix.ts | Cross | EXIST — expand | ~1,800 |
| strRegulationDb.ts | Cross | NEW | ~2,400 |
| commandCenterUi.tsx | Cross | NEW | ~4,500 |

**END OF ULTRAPLAN — BEGIN EXECUTION**

The doctrine is set. The diagnostic is honest. The roadmap is sequenced. The KPIs are calibrated. The code modules are inventoried. What remains is execution — phase by phase, module by module, file by file. The platform will not become elite because of this document; it will become elite because the operator desk lives the doctrine every day, declines the bad files, sends the clean files, and tells the borrower the truth in ten minutes. Build the doctrine. Live the doctrine. Win the market.